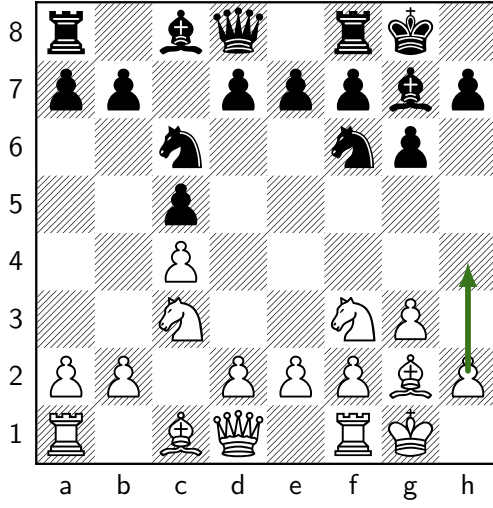


Melanie-ANALOGY — English, Symmetrical

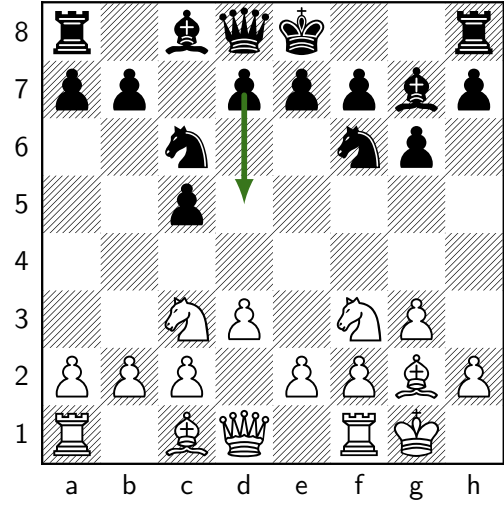
Melanie (Deep_Vision)

Abstract

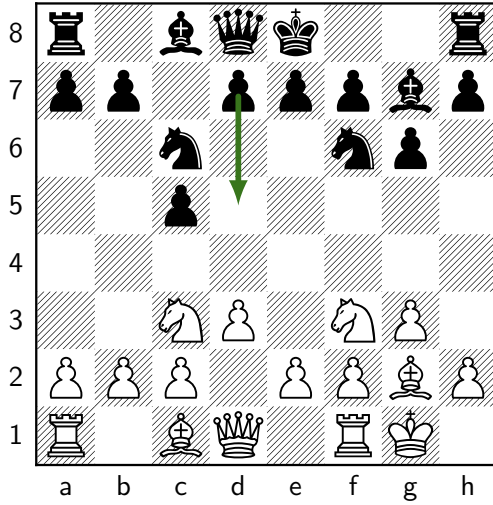
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



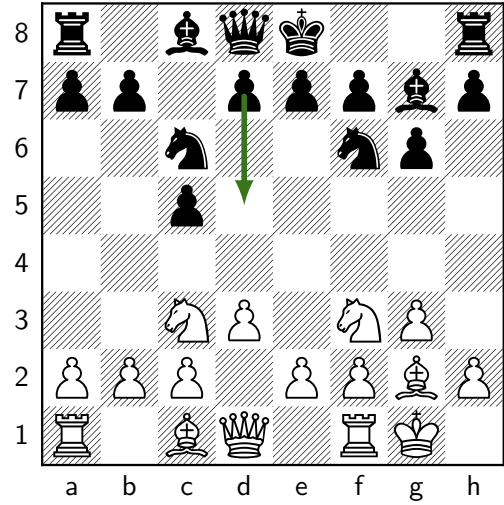
(a)



(b)

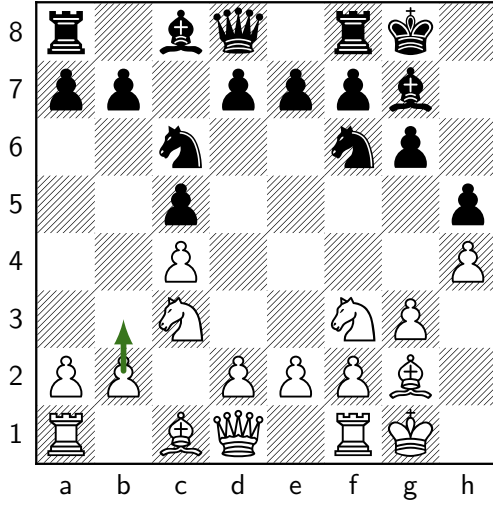


(c)

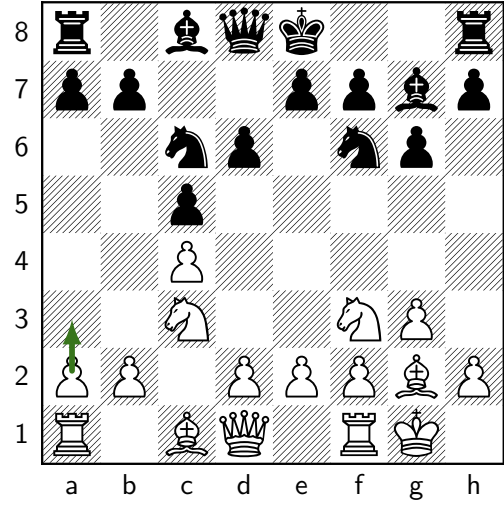


(d)

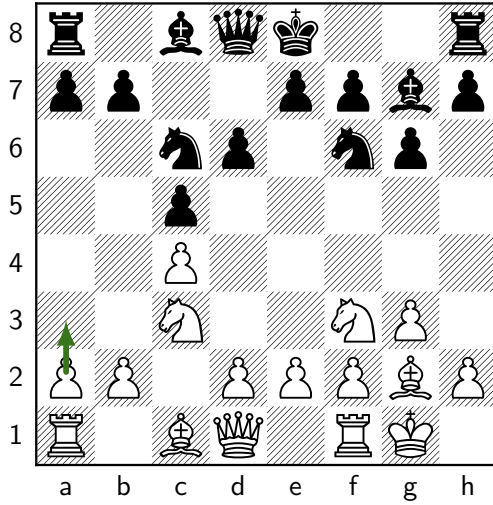
Figure 1: **Move 7, White: h4.** (a) Our position; Melanie plays **h4** (**h2h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#542` (ply 11) [structural distance $d=31$, QAP obj=8]: the master played **d5**; under μ our *Ph2* plays the role of that game’s *Pd2*; cross-type roles: our *Pc4* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **h2h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#543` (ply 11) [structural distance $d=31$, QAP obj=8]: the master played **d5**; under μ our *Ph2* plays the role of that game’s *Pd2*; cross-type roles: our *Pc4* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **h2h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1424` (ply 11) [structural distance $d=31$, QAP obj=8]: the master played **d5**; under μ our *Ph2* plays the role of that game’s *Pd2*; cross-type roles: our *Pc4* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **h2h4** (**endorses this move**). Synthesis: 10 retrieved analogs transferred to **h4** (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



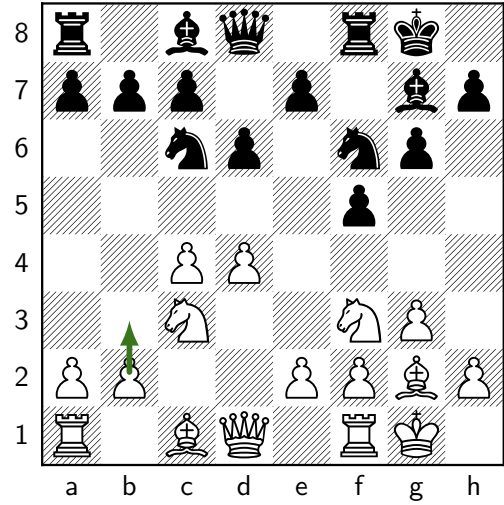
(a)



(b)

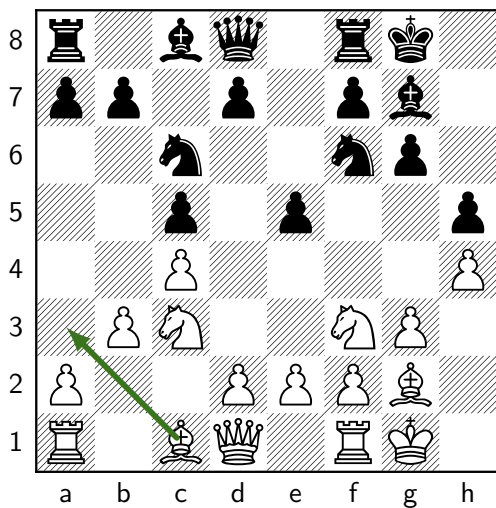


(c)

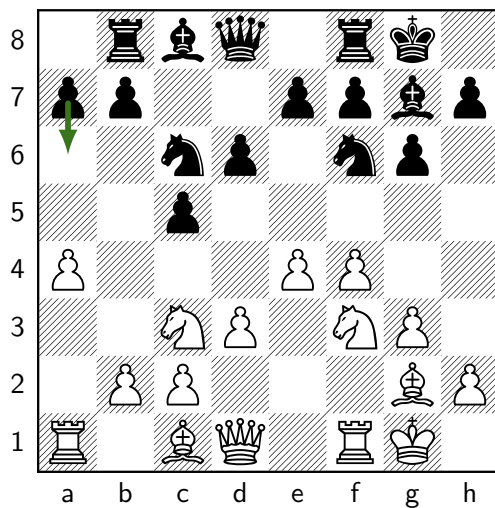


(d)

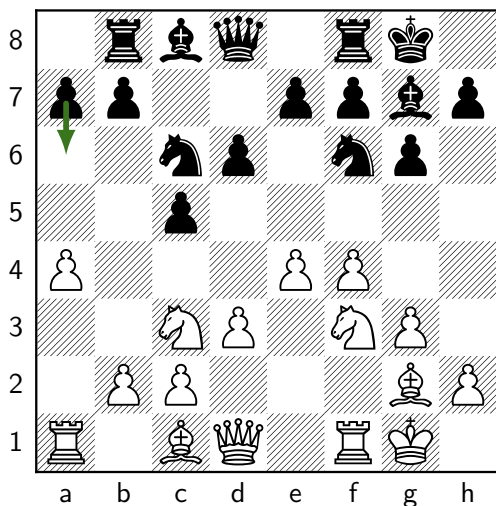
Figure 2: **Move 8, White: b3**. (a) Our position; Melanie plays **b3** (**b2b3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#870` (ply 12) [structural distance $d=29$, QAP obj=8]: the master played **a3**; under μ our *Pb2* plays the role of that game’s *Pa2*; cross-type roles: our *Bg2* as their *Pf2*, our *Pf2* as their *Bg2*; transferred move **b2b3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#871` (ply 12) [structural distance $d=29$, QAP obj=8]: the master played **a3**; under μ our *Pb2* plays the role of that game’s *Pa2*; cross-type roles: our *Bg2* as their *Pf2*, our *Pf2* as their *Bg2*; transferred move **b2b3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3573` (ply 14) [structural distance $d=22$, QAP obj=7]: the master played **b3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Ph4* as their *Ra1*, our *nf6* as their *Bc1*; transferred move **b2b3** (**endorses this move**). Synthesis: 10 retrieved analogs transferred to **b3** (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



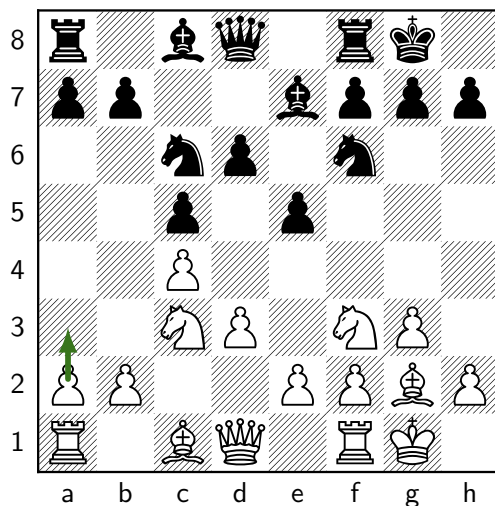
(a)



(b)

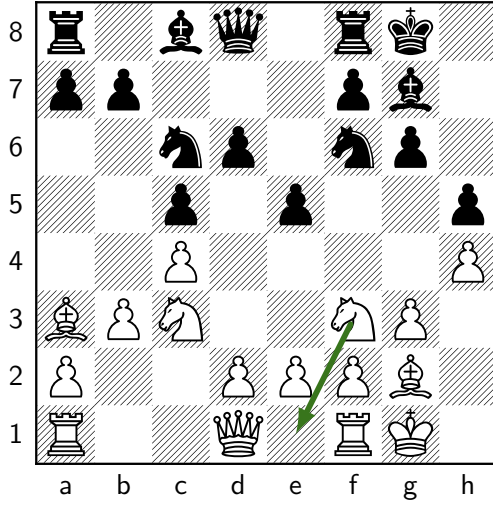


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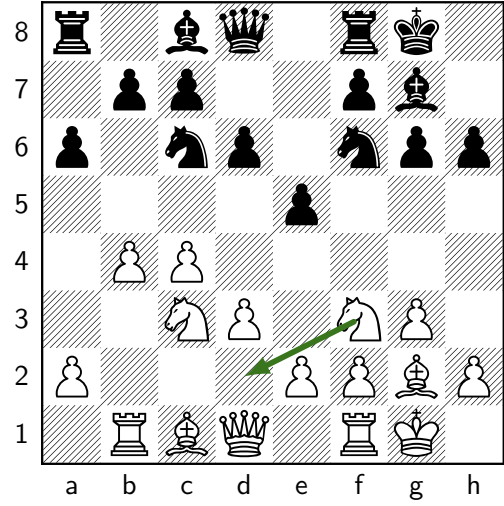


(d)

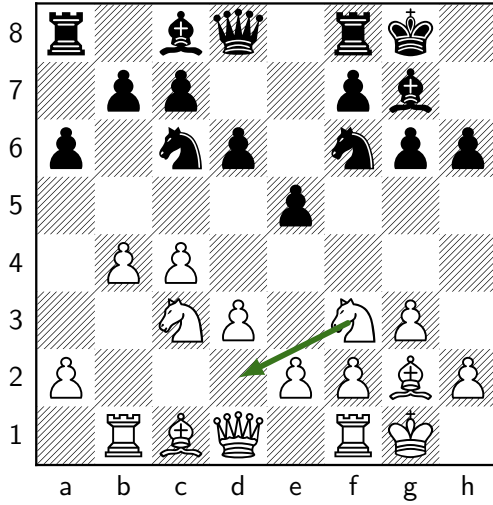
Figure 3: **Move 9, White: Ba3.** (a) Our position; Melanie plays Ba3 (c1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2492` (ply 17) [structural distance $d=27$, QAP obj=35]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *nf6* as their *Rb8*, our *Pg3* as their *Bc8*; transferred move c1a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2593` (ply 17) [structural distance $d=27$, QAP obj=35]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *nf6* as their *Rb8*, our *Pg3* as their *Bc8*; transferred move c1a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#794` (ply 14) [structural distance $d=23$, QAP obj=33]: the master played a3; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *pb7* as their *Bc1*, our *Pb3* as their *Qd1*; transferred move c1a3 (**endorses this move**). Synthesis: 26 retrieved analogs transferred to Ba3 (summed QAP objective 694, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



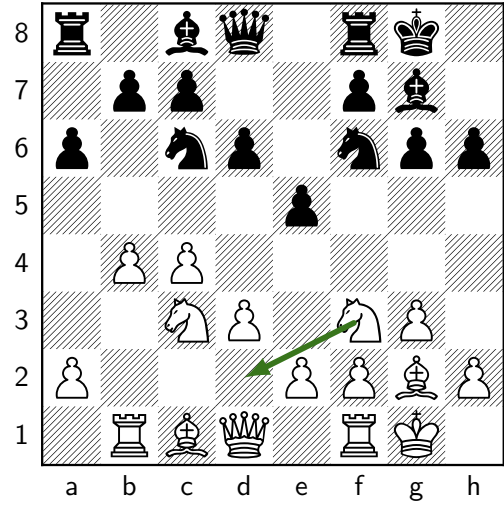
(a)



(b)

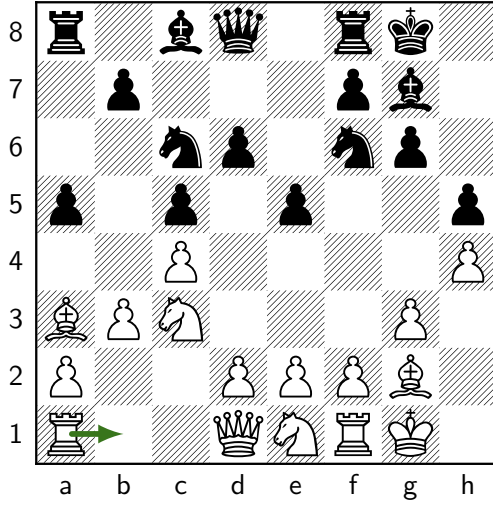


(c)

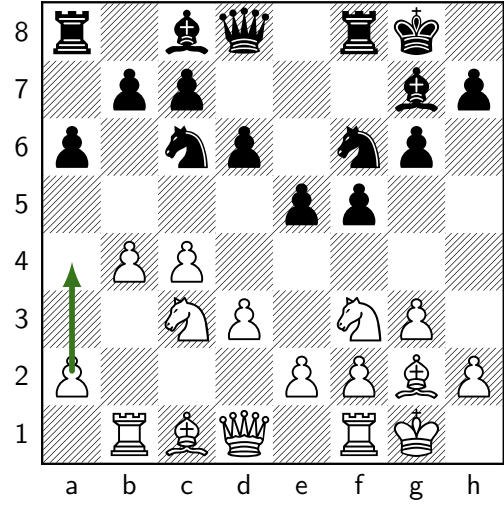


(d)

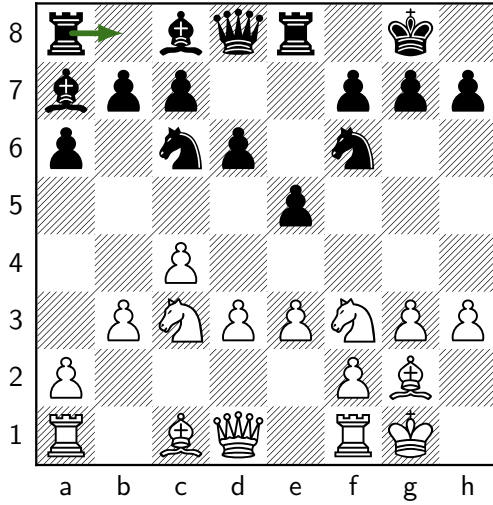
Figure 4: **Move 10, White: Ne1**. (a) Our position; Melanie plays Ne1 (f3e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1169` (ply 18) [structural distance $d=37$, QAP obj=46]: the master played Nd2; under μ our Nf3 plays the role of that game's Nf3; cross-type roles: our Pa2 as their Rb1, our pg6 as their Bc1; transferred move f3e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1173` (ply 18) [structural distance $d=37$, QAP obj=46]: the master played Nd2; under μ our Nf3 plays the role of that game's Nf3; cross-type roles: our Pa2 as their Rb1, our pg6 as their Bc1; transferred move f3e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1174` (ply 18) [structural distance $d=37$, QAP obj=46]: the master played Nd2; under μ our Nf3 plays the role of that game's Nf3; cross-type roles: our Pa2 as their Rb1, our pg6 as their Bc1; transferred move f3e1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ne1 (summed QAP objective 217, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



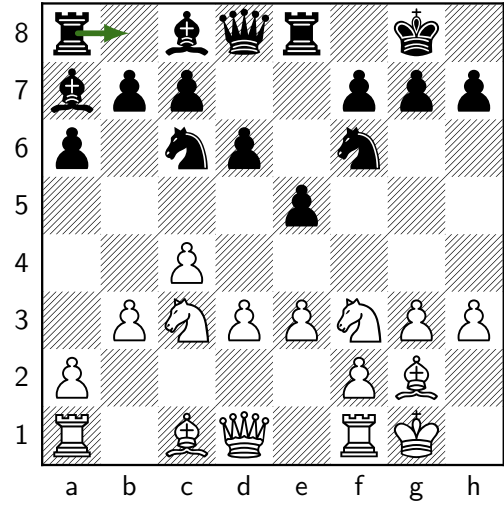
(a)



(b)

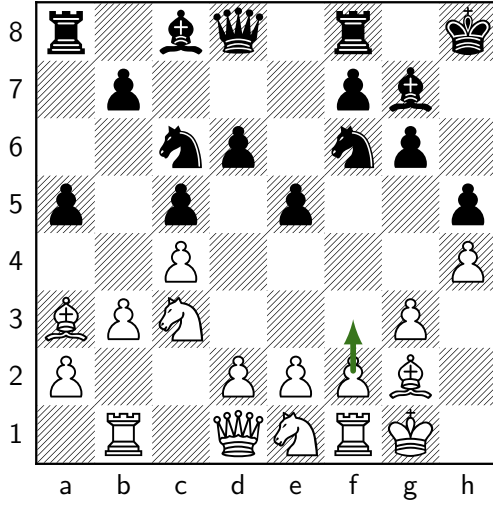


(c)

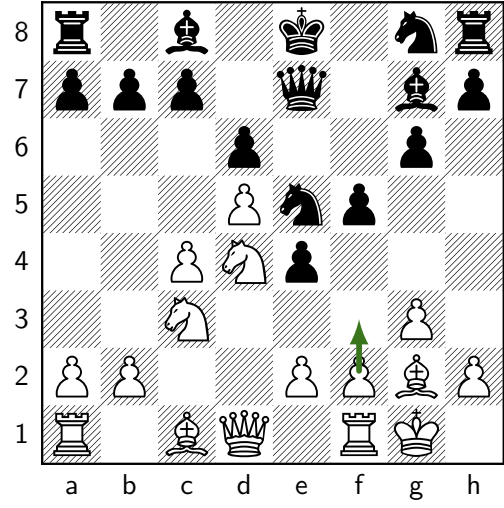


(d)

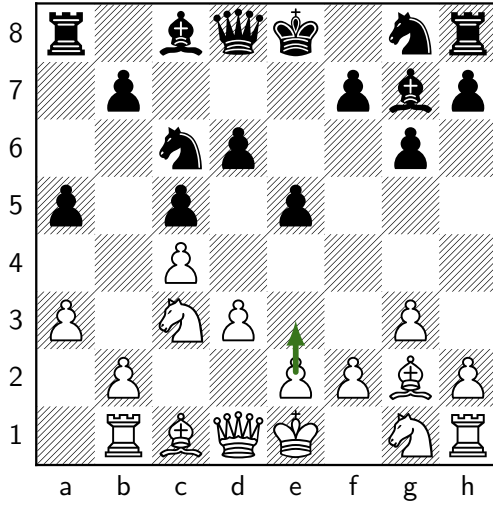
Figure 5: **Move 11, White: Rb1.** (a) Our position; Melanie plays Rb1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3982` (ply 18) [structural distance $d=51$, QAP obj=32]: the master played a4; under μ our *Ra1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *pa5* as their *Rb1*, our *pb7* as their *Bc1*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2943` (ply 19) [structural distance $d=51$, QAP obj=24]: the master played Rb8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *nf6* as their *Bc8*, our *Pg3* as their *Qd8*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2944` (ply 19) [structural distance $d=51$, QAP obj=24]: the master played Rb8; under μ our *Ra1* plays the role of that game’s *Ra1*; cross-type roles: our *nf6* as their *Bc8*, our *Pg3* as their *Qd8*; transferred move a1b1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rb1 (summed QAP objective 171, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



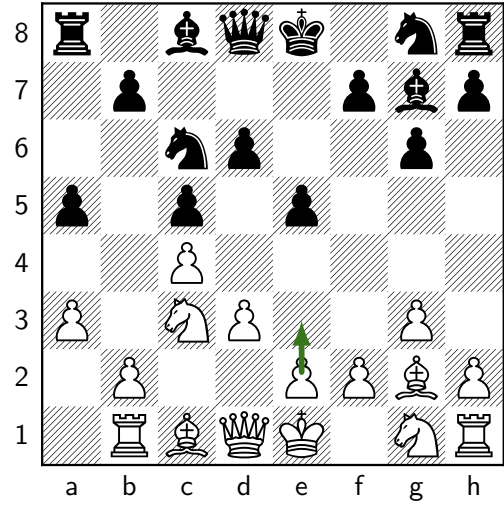
(a)



(b)

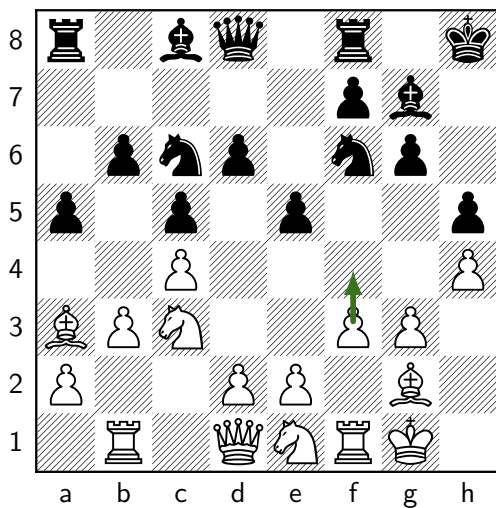


(c)

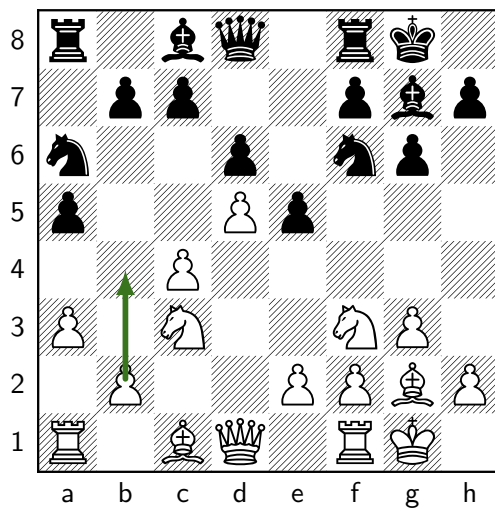


(d)

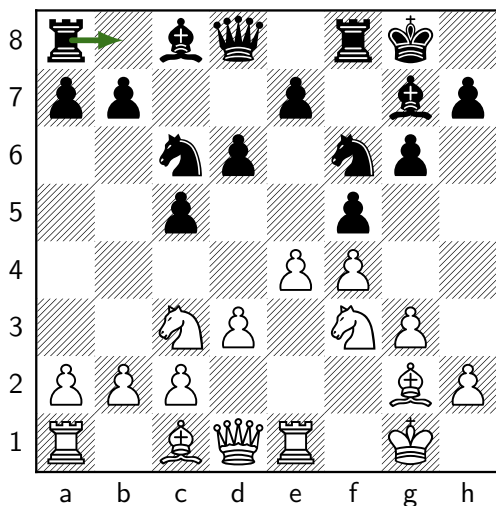
Figure 6: **Move 12, White: f3**. (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#917` (ply 18) [structural distance $d=60$, QAP obj=44]: the master played f3; under μ our $Pf2$ plays the role of that game’s $Pf2$; cross-type roles: our $ph5$ as their $Bc1$, our $Pb3$ as their $Qd1$; transferred move f2f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1278` (ply 14) [structural distance $d=60$, QAP obj=28]: the master played e3; under μ our $Pf2$ plays the role of that game’s $Pe2$; cross-type roles: our $Pg3$ as their $Rb1$, our $nf6$ as their $Bc1$; transferred move f2f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1279` (ply 14) [structural distance $d=60$, QAP obj=28]: the master played e3; under μ our $Pf2$ plays the role of that game’s $Pe2$; cross-type roles: our $Pg3$ as their $Rb1$, our $nf6$ as their $Bc1$; transferred move f2f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to f3 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



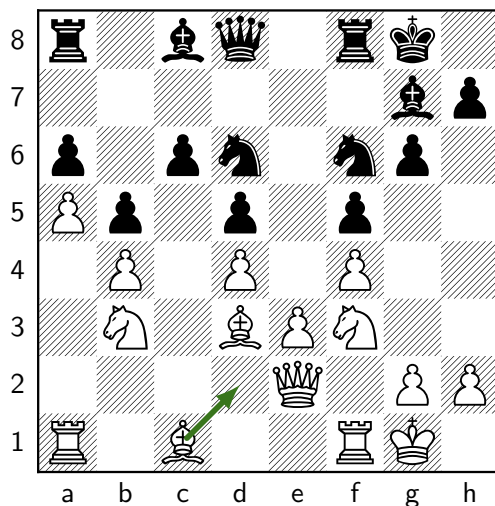
(a)



(b)

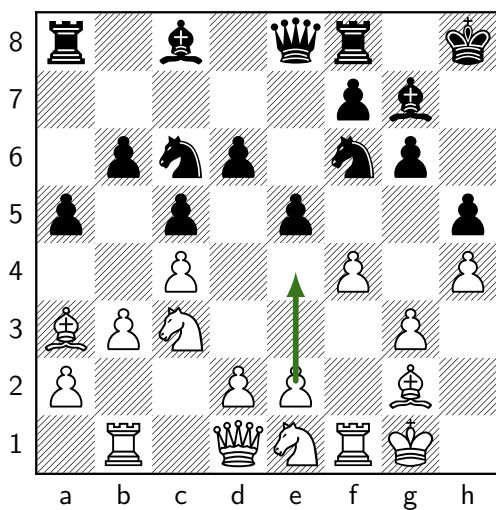


(c)

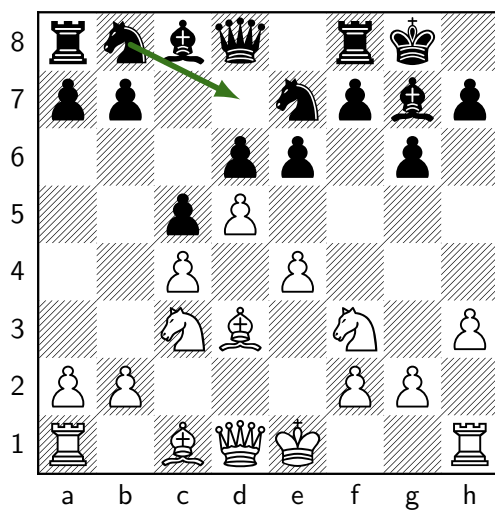


(d)

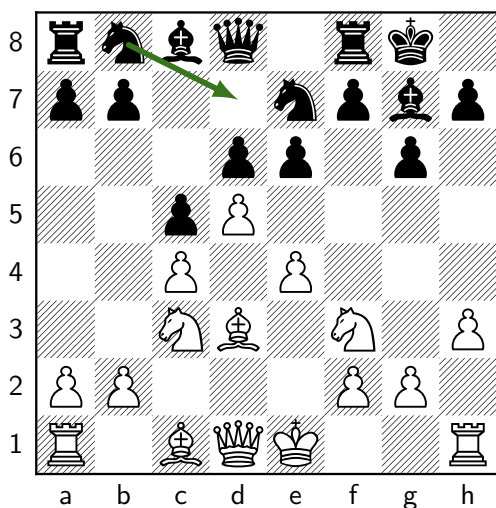
Figure 7: **Move 13, White: f4**. (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1091` (ply 22) [structural distance $d=75$, QAP obj=27]: the master played b4; under μ our *Pf3* plays the role of that game’s *Pb2*; cross-type roles: our *Qd1* as their *Ra1*, our *Nc3* as their *Bc1*; transferred move f3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2609` (ply 17) [structural distance $d=71$, QAP obj=23]: the master played Rb8; under μ our *Pf3* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Pb3* as their *Bc8*, our *Nc3* as their *Qd8*; transferred move f3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1758` (ply 36) [structural distance $d=73$, QAP obj=47]: the master played Bd2; under μ our *Ne1* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Ph4* as their *Ra1*, our *Pa2* as their *Qe2*; transferred move e1c2. Synthesis: 2 retrieved analogs transferred to f4 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



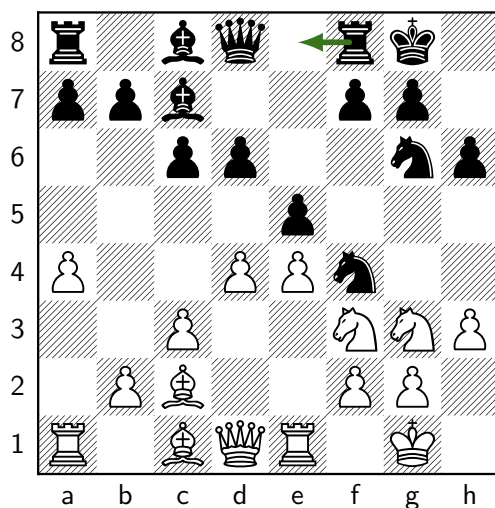
(a)



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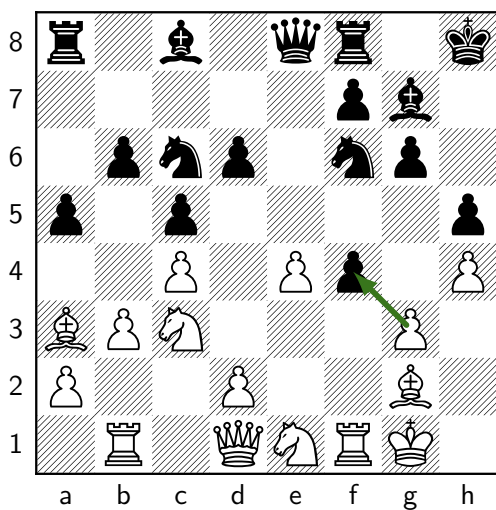


(c)

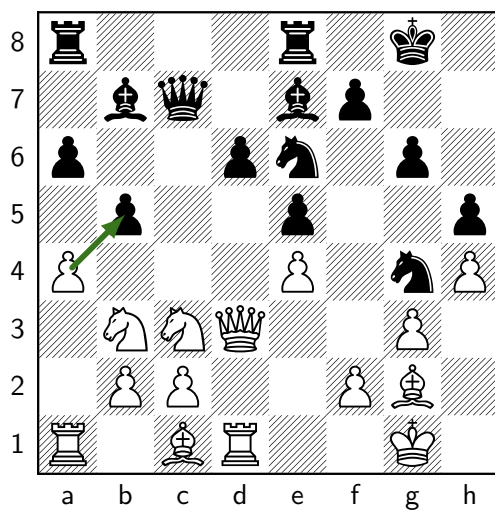


(d)

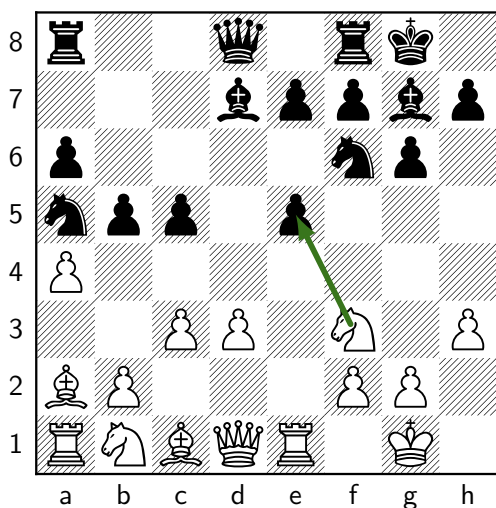
Figure 8: **Move 14, White: e4**. (a) Our position; Melanie plays e4 (e2e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3750` (ply 15) [structural distance $d=81$, QAP obj=70]: the master played Nd7; under μ our *Pe2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pb3* as their *Ra8*, our *pb6* as their *Bc8*; transferred move e2e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3751` (ply 15) [structural distance $d=81$, QAP obj=70]: the master played Nd7; under μ our *Pe2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pb3* as their *Ra8*, our *pb6* as their *Bc8*; transferred move e2e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1206` (ply 29) [structural distance $d=77$, QAP obj=70]: the master played Re8; under μ our *Pe2* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Ra8*, our *Pc4* as their *Bc8*; transferred move e2e4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to e4 (summed QAP objective 211, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



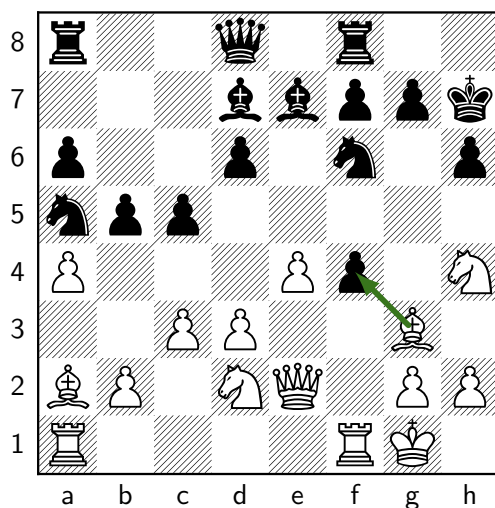
(a)



(b)

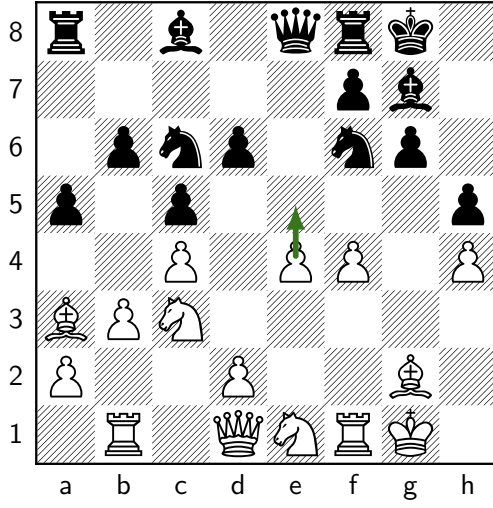


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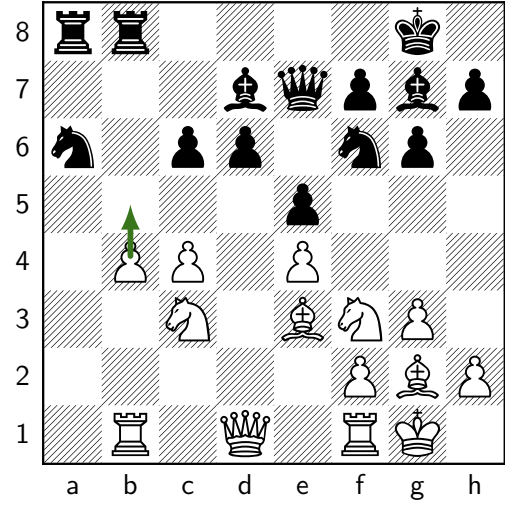


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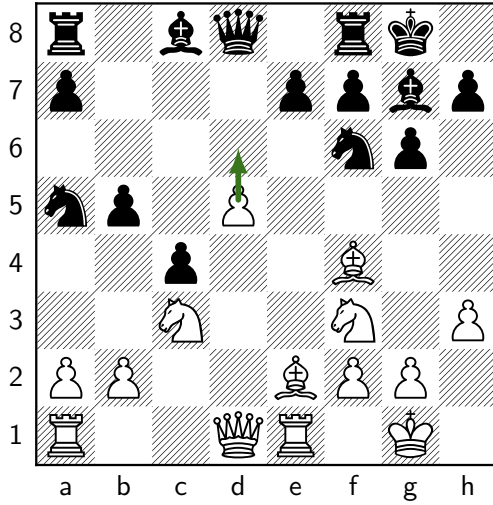
Figure 9: **Move 15, White: gxf4.** (a) Our position; Melanie plays gxf4 (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#1011 (ply 38) [structural distance $d=82$, QAP obj=111]: the master played axb5; under μ our $Pg3$ plays the role of that game’s $Pa4$; cross-type roles: our $Bg2$ as their $Ra1$, our $pb6$ as their $Bc1$; transferred move g3f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#178 (ply 24) [structural distance $d=84$, QAP obj=73]: the master played Nxe5; under μ our $Pg3$ plays the role of that game’s $Nf3$ (a **cross-type** correspondence); cross-type roles: our $pf7$ as their $Nb1$, our $Ne1$ as their $Bc1$; transferred move g3f4 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3565 (ply 30) [structural distance $d=73$, QAP obj=66]: the master played Bxf4; under μ our $Pg3$ plays the role of that game’s $Bg3$ (a **cross-type** correspondence); cross-type roles: our $pa5$ as their $Ra1$, our $pb6$ as their $Ba2$; transferred move g3f4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to gxf4 (summed QAP objective 405, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



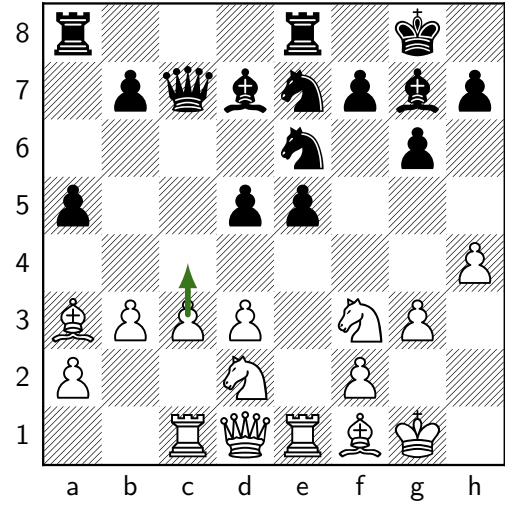
(a)



(b)

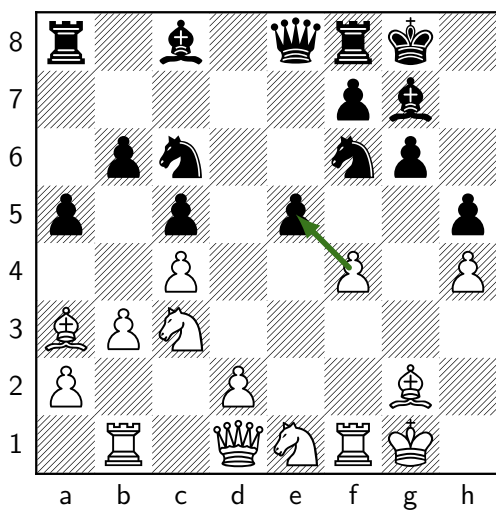


(c)

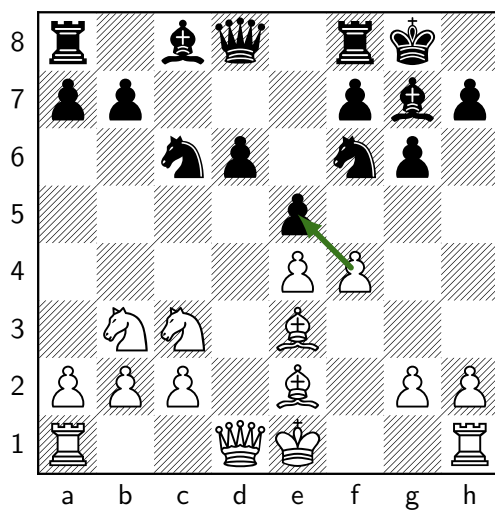


(d)

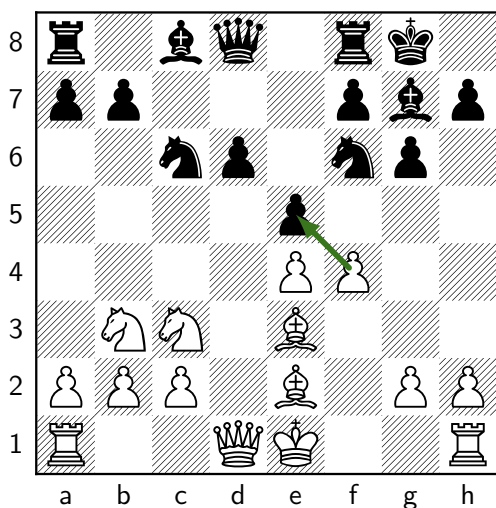
Figure 10: **Move 16, White: e5.** (a) Our position; Melanie plays e5 (e4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1091` (ply 38) [structural distance $d=72$, QAP obj=115]: the master played b5; under μ our Pe_4 plays the role of that game’s Pb_4 ; cross-type roles: our Nc_3 as their Rb_1 , our bg_7 as their Pf_2 ; transferred move e4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3742` (ply 24) [structural distance $d=74$, QAP obj=99]: the master played d6; under μ our Pe_4 plays the role of that game’s Pd_5 ; cross-type roles: our Pc_4 as their Ra_1 , our Bg_2 as their Qd_1 ; transferred move e4e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1809` (ply 30) [structural distance $d=67$, QAP obj=90]: the master played c4; under μ our Pe_4 plays the role of that game’s Pc_3 ; cross-type roles: our Nc_3 as their Rc_1 , our Bg_2 as their Qd_1 ; transferred move e4e5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to e5 (summed QAP objective 378, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



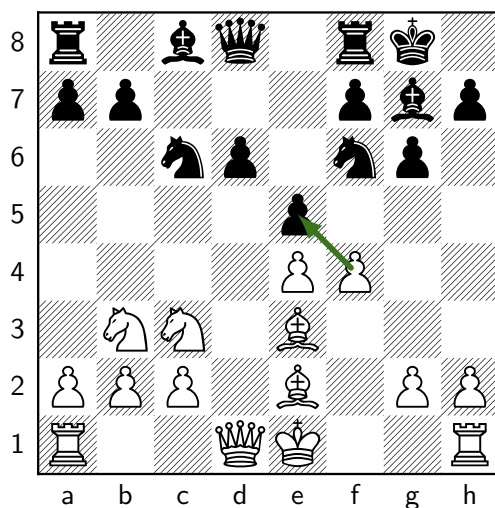
(a)



(b)

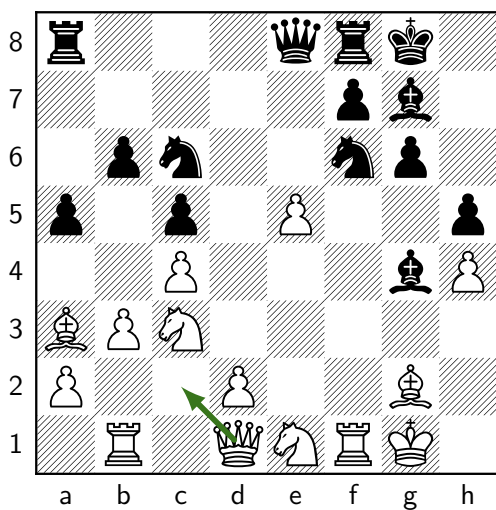


(c)

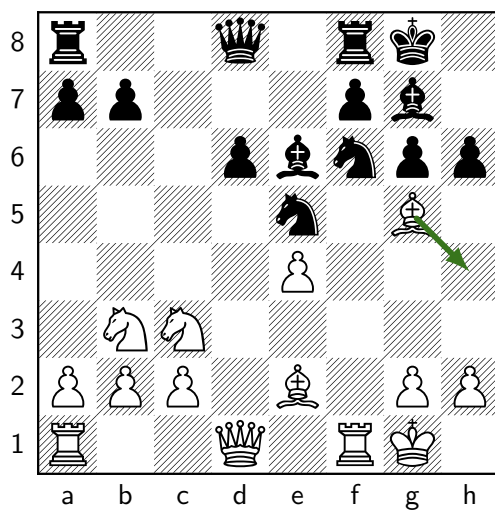


(d)

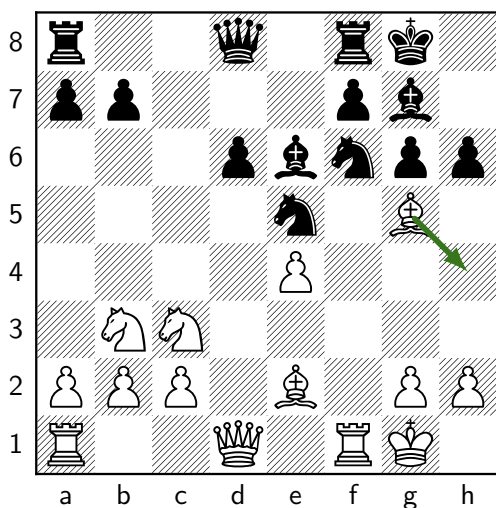
Figure 11: **Move 17, White: fxe5.** (a) Our position; Melanie plays fxe5 (f4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#1319 (ply 18) [structural distance $d=62$, QAP obj=66]: the master played fxe5; under μ our Pf4 plays the role of that game’s Pf4; cross-type roles: our Ne1 as their Ra1, our Bg2 as their Qd1; transferred move f4e5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1320 (ply 18) [structural distance $d=62$, QAP obj=66]: the master played fxe5; under μ our Pf4 plays the role of that game’s Pf4; cross-type roles: our Ne1 as their Ra1, our Bg2 as their Qd1; transferred move f4e5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2327 (ply 18) [structural distance $d=62$, QAP obj=66]: the master played fxe5; under μ our Pf4 plays the role of that game’s Pf4; cross-type roles: our Ne1 as their Ra1, our Bg2 as their Qd1; transferred move f4e5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to fxe5 (summed QAP objective 667, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



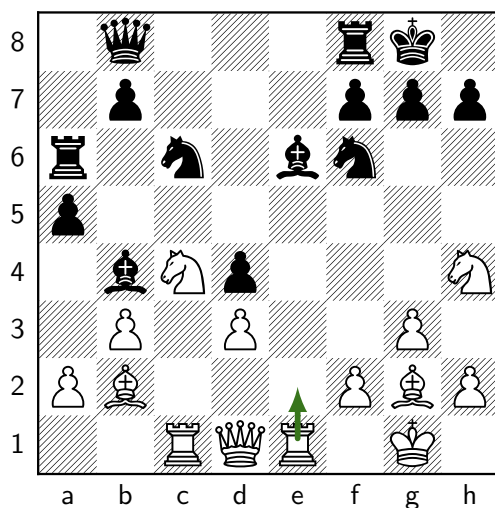
(a)



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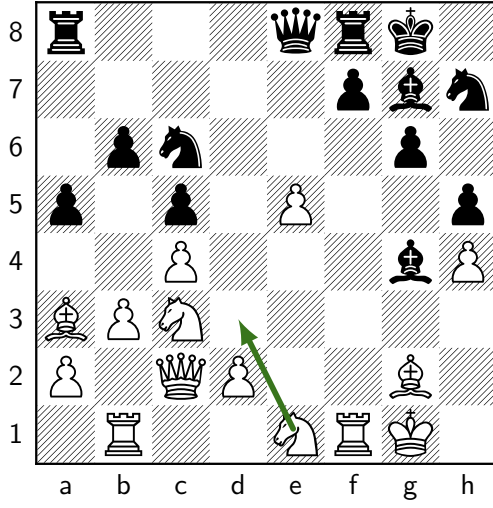


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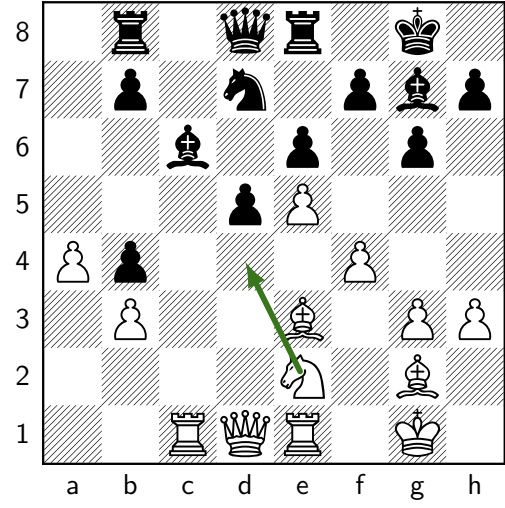


(d)

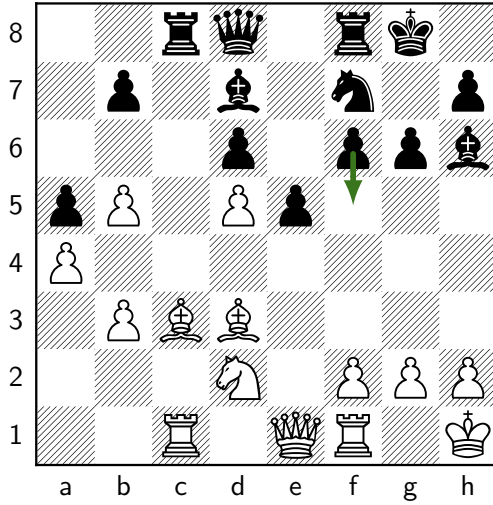
Figure 12: **Move 18, White: Qc2.** (a) Our position; Melanie plays Qc2 (d1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1320` (ply 24) [structural distance $d=84$, QAP obj=42]: the master played Bh4; under μ our Qd1 plays the role of that game's Bg5 (a **cross-type** correspondence); cross-type roles: our ph5 as their Ra1, our Bg2 as their Qd1; transferred move d1c2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2946` (ply 24) [structural distance $d=84$, QAP obj=42]: the master played Bh4; under μ our Qd1 plays the role of that game's Bg5 (a **cross-type** correspondence); cross-type roles: our ph5 as their Ra1, our Bg2 as their Qd1; transferred move d1c2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1080` (ply 32) [structural distance $d=93$, QAP obj=41]: the master played Re2; under μ our Qd1 plays the role of that game's Re1 (a **cross-type** correspondence); cross-type roles: our bg7 as their Qd1, our Ne1 as their Pa2; transferred move d1c2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qc2 (summed QAP objective 190, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



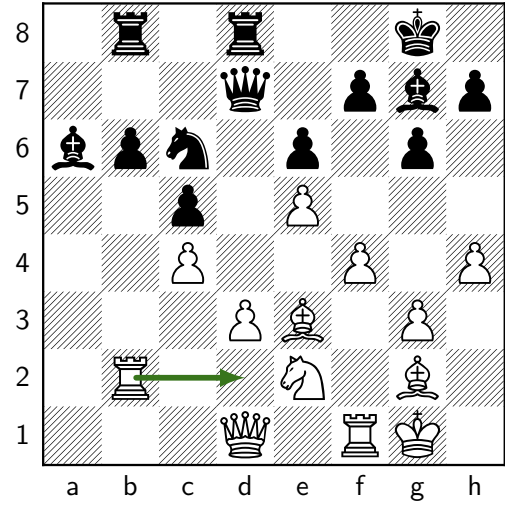
(a)



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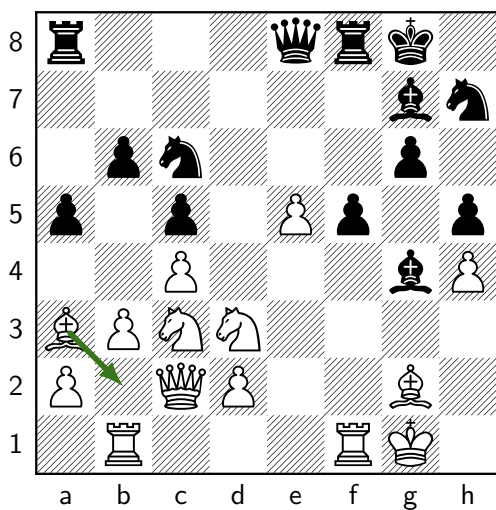


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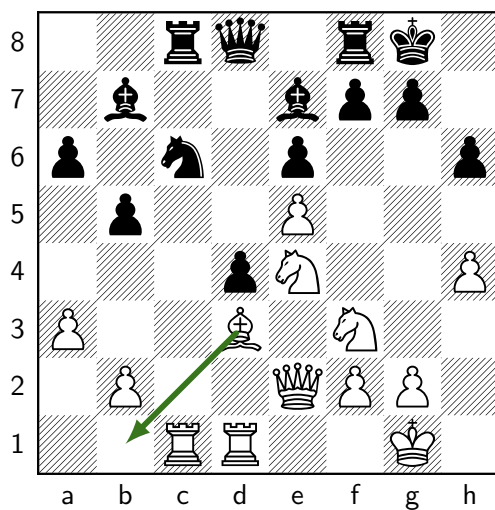


(d)

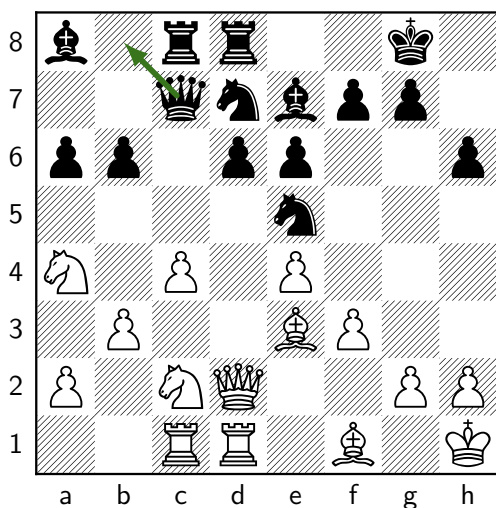
Figure 13: **Move 19, White: Nd3.** (a) Our position; Melanie plays Nd3 (e1d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1695` (ply 46) [structural distance $d=89$, QAP obj=113]: the master played Nd4; under μ our *Ne1* plays the role of that game’s *Ne2*; cross-type roles: our *Bg2* as their *Rc1*, our *nh7* as their *Pb3*; transferred move e1d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2435` (ply 49) [structural distance $d=90$, QAP obj=184]: the master played f5; under μ our *Nc3* plays the role of that game’s *Pf3* (a **cross-type** correspondence); cross-type roles: our *Bg2* as their *Rc8*, our *Ne1* as their *Qd8*; transferred move c3e4. (d) Retrieved from `ccr1_top_p0001#3912` (ply 38) [structural distance $d=92$, QAP obj=127]: the master played Rd2; under μ our *Qc2* plays the role of that game’s *Rb2* (a **cross-type** correspondence); cross-type roles: our *pa5* as their *Qd1*, our *pb6* as their *Rf1*; transferred move c2b2. Synthesis: 1 retrieved analog transferred to Nd3 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



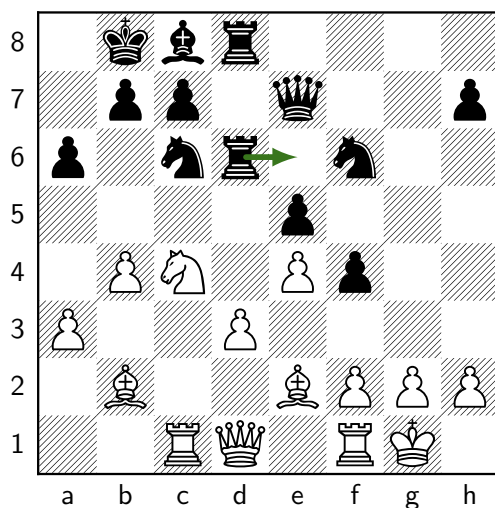
(a)



(b)

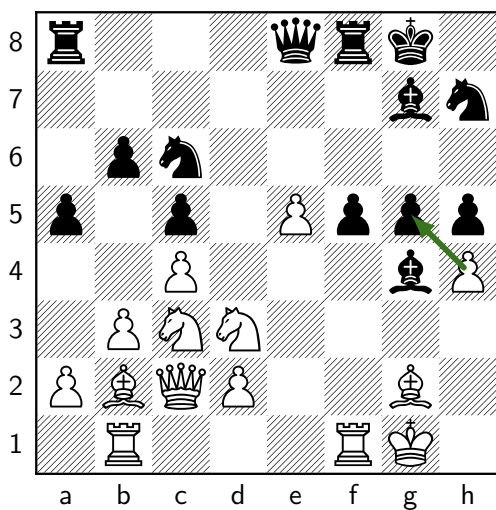


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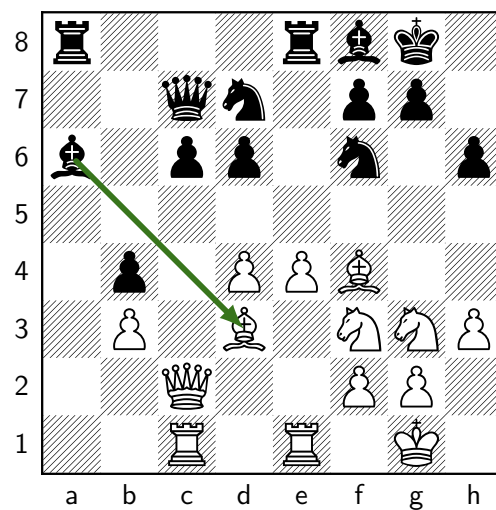


(d)

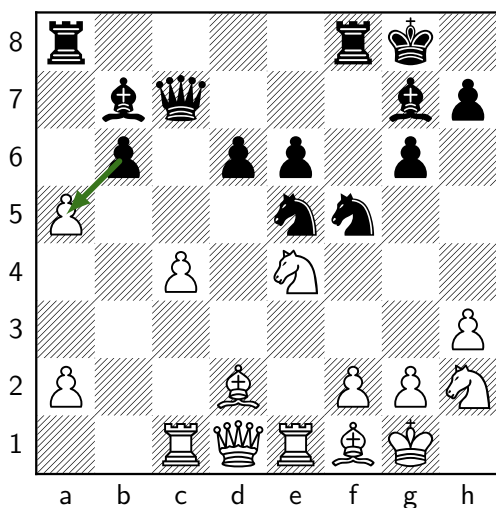
Figure 14: **Move 20, White: Bb2.** (a) Our position; Melanie plays Bb2 (a3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#720` (ply 34) [structural distance $d=92$, QAP obj=109]: the master played Bb1; under μ our Ba3 plays the role of that game’s Bd3; cross-type roles: our Bg2 as their Rc1, our Pd2 as their Qe2; transferred move a3b2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#561` (ply 43) [structural distance $d=93$, QAP obj=87]: the master played Qb8; under μ our Ba3 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Rf1 as their Ba8, our Bg2 as their Rc8; transferred move a3b2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3599` (ply 39) [structural distance $d=89$, QAP obj=33]: the master played Re6; under μ our Ba3 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Rf1 as their Bc8, our Pa2 as their Rd8; transferred move a3b2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bb2 (summed QAP objective 229, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



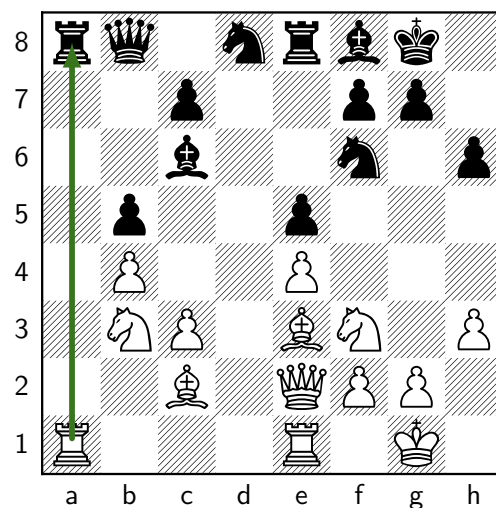
(a)



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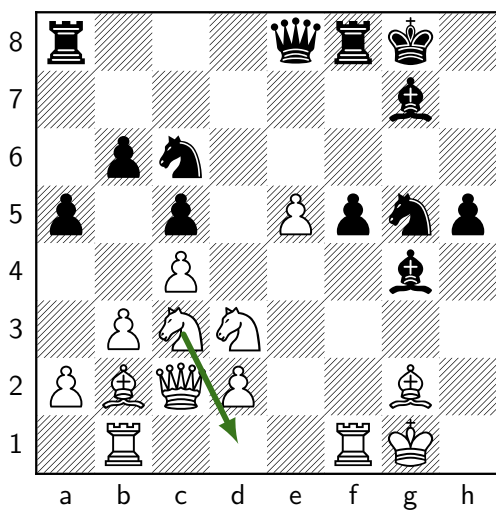


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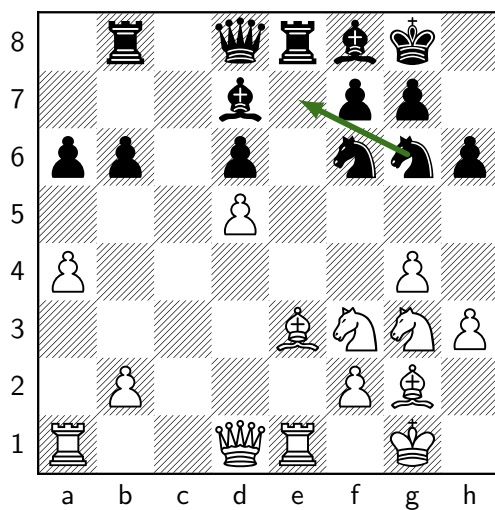


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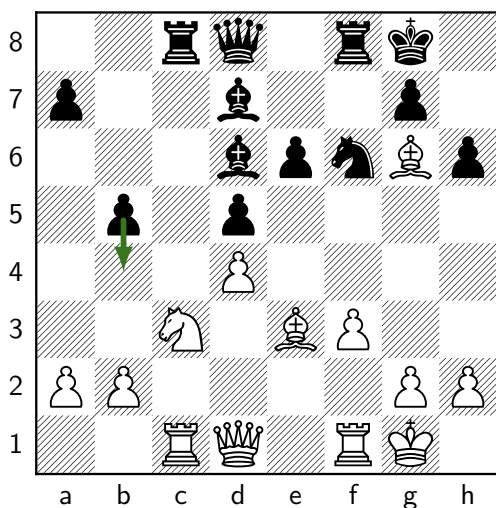
Figure 15: **Move 21, White: hxg5.** (a) Our position; Melanie plays hxg5 (h4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#84 (ply 45) [structural distance $d=88$, QAP obj=97]: the master played Bxd3; under μ our Ph_4 plays the role of that game’s $Ba3$ (a **cross-type** correspondence); cross-type roles: our $ra8$ as their $Qc7$, our $Nd3$ as their $Pg7$; transferred move h4g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3251 (ply 39) [structural distance $d=92$, QAP obj=88]: the master played bxa5; under μ our Ph_4 plays the role of that game’s $Pb3$; cross-type roles: our Ph_4 as their $Ra8$, our $bg7$ as their $Qc7$; transferred move h4g5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#2079 (ply 36) [structural distance $d=77$, QAP obj=77]: the master played Rxa8; under μ our Ph_4 plays the role of that game’s $Ra1$ (a **cross-type** correspondence); cross-type roles: our $Bb2$ as their $Re1$, our $Pa2$ as their $Bc2$; transferred move h4g5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to hxg5 (summed QAP objective 579, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



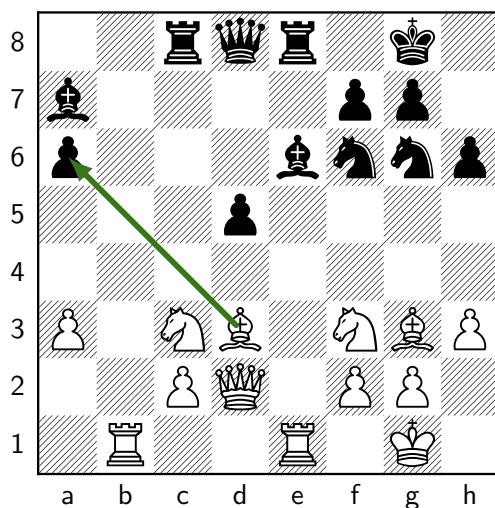
(a)



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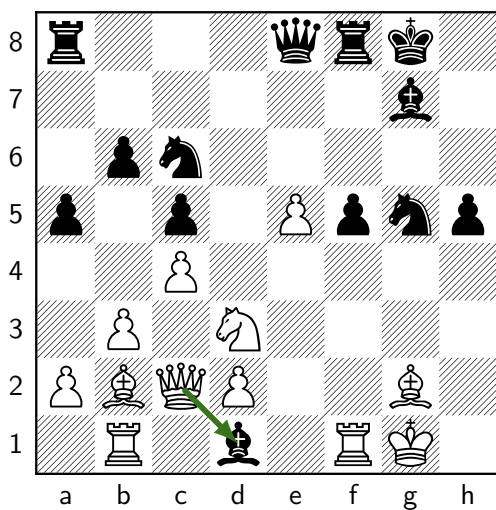


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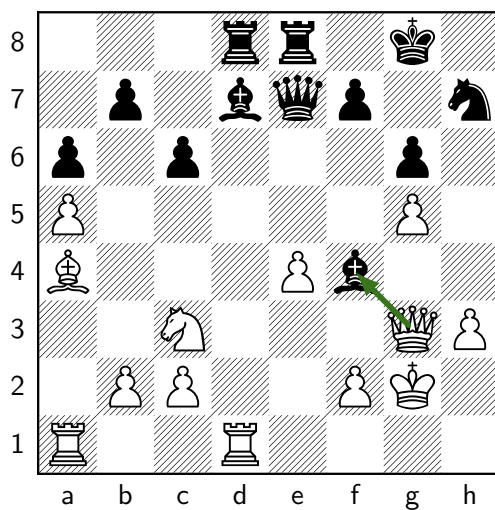


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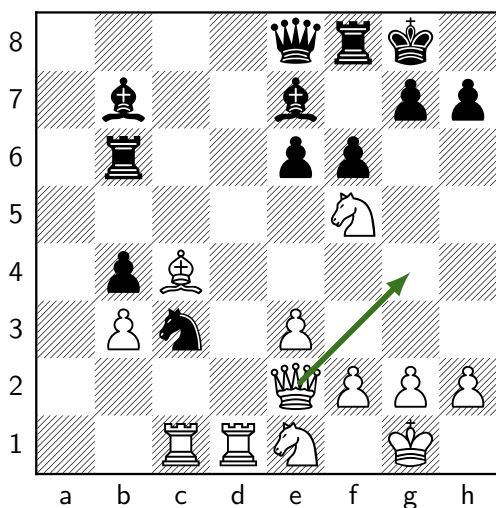
Figure 16: **Move 22, White: Nd1.** (a) Our position; Melanie plays Nd1 (c3d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0002#3019` (ply 41) [structural distance $d=93$, QAP obj=149]: the master played Ne7; under μ our Nc3 plays the role of that game’s Ng3; cross-type roles: our Rf1 as their Qd8, our Bg2 as their Re8; transferred move c3d1 (**endorses this move**). (c) Retrieved from `ccr1.top_p0000#410` (ply 35) [structural distance $d=98$, QAP obj=140]: the master played b4; under μ our Pe5 plays the role of that game’s Pb4; cross-type roles: our Bg2 as their Rc8, our Bb2 as their Qd8; transferred move e5e6. (d) Retrieved from `ccr1.top_p0000#3591` (ply 44) [structural distance $d=95$, QAP obj=113]: the master played Bxa6; under μ our Nd3 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Bg2 as their Re1, our Bb2 as their Pc2; transferred move d3c5. Synthesis: 2 retrieved analogs transferred to Nd1 (summed QAP objective 298, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



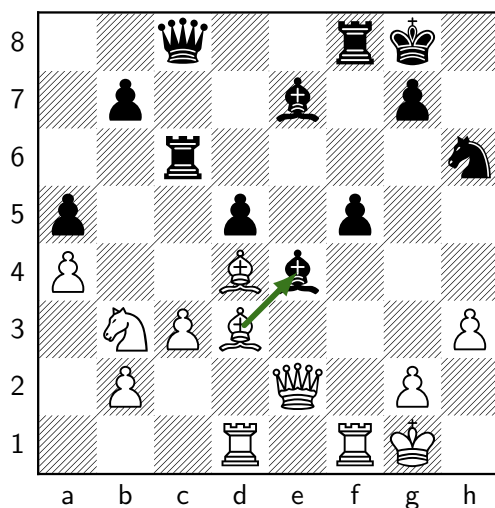
(a)



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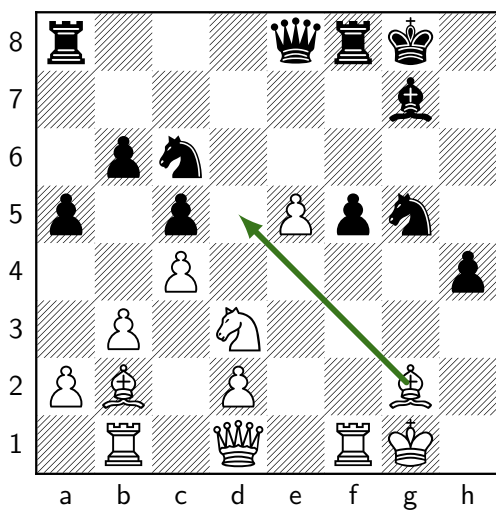


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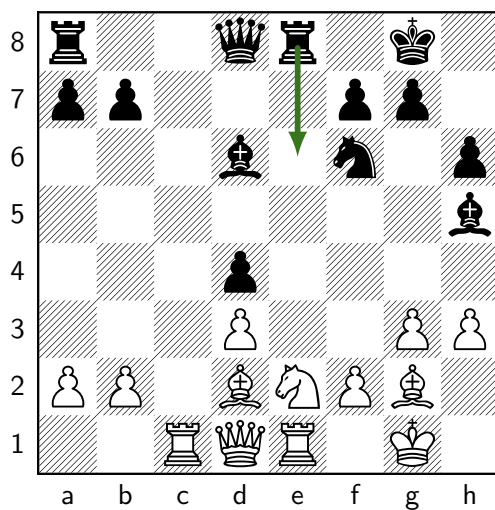


(d)

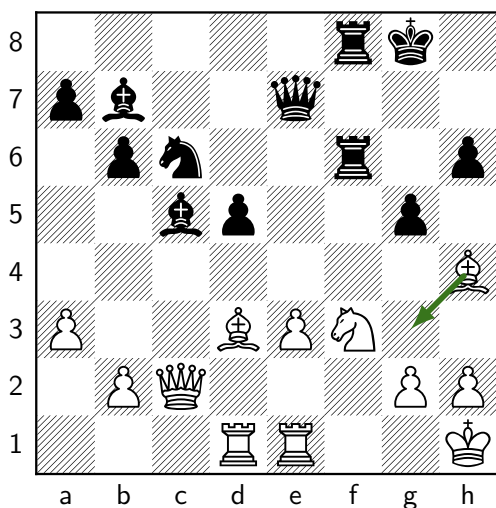
Figure 17: **Move 23, White: Qxd1.** (a) Our position; Melanie plays Qxd1 (c2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2191 (ply 54) [structural distance $d=82$, QAP obj=67]: the master played Qxf4; under μ our Qc2 plays the role of that game's Qg3; cross-type roles: our Bg2 as their Rd1, our Rb1 as their Pb2; transferred move c2d1 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#3362 (ply 58) [structural distance $d=99$, QAP obj=56]: the master played Qg4; under μ our Qc2 plays the role of that game's Qe2; cross-type roles: our Qc2 as their Rd1, our Rf1 as their Ph2; transferred move c2d1 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1284 (ply 52) [structural distance $d=97$, QAP obj=50]: the master played Bxe4; under μ our Qc2 plays the role of that game's Bd3 (a **cross-type** correspondence); cross-type roles: our Bg2 as their Rd1, our Rb1 as their Qe2; transferred move c2d1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qxd1 (summed QAP objective 302, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



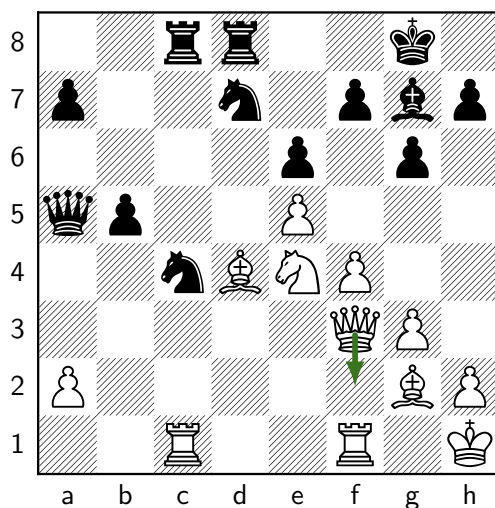
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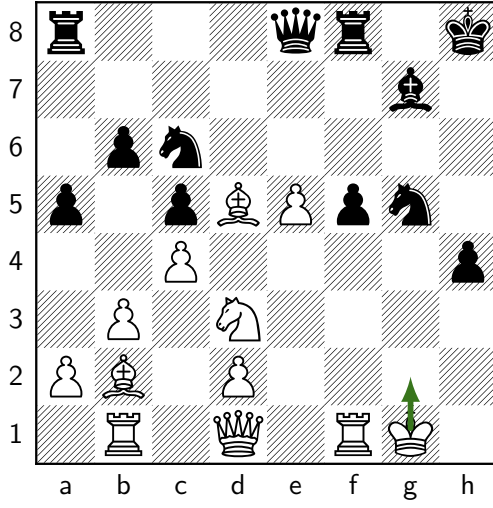


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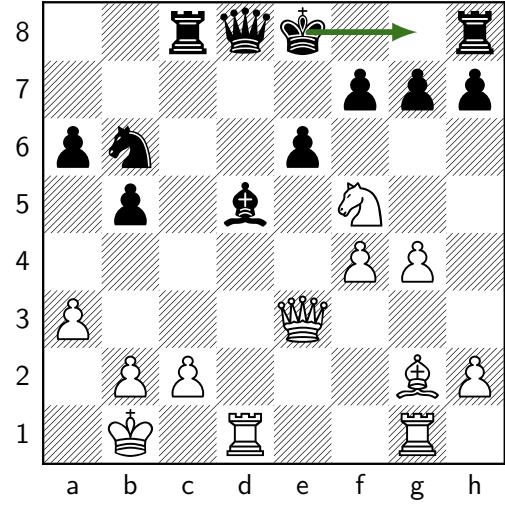


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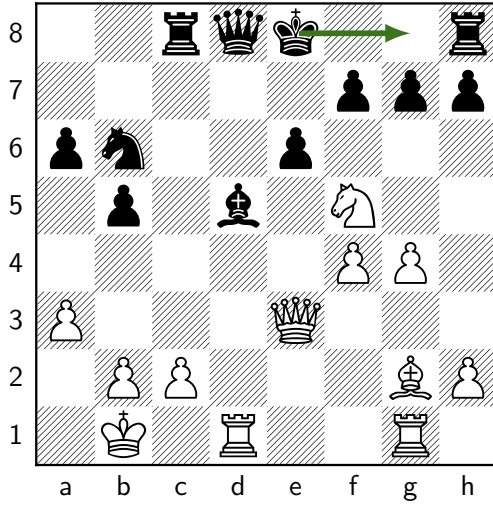
Figure 18: **Move 24, White: Bd5+.** (a) Our position; Melanie plays Bd5+ (g2d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1084` (ply 33) [structural distance $d=73$, QAP obj=122]: the master played Re6; under μ our Bg2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Rf1 as their Pa7, our Bb2 as their Pg7; transferred move g2d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1185` (ply 46) [structural distance $d=67$, QAP obj=31]: the master played Bg3; under μ our Bg2 plays the role of that game’s Bh4; cross-type roles: our ng5 as their Rd1, our Bb2 as their Re1; transferred move g2d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2998` (ply 44) [structural distance $d=63$, QAP obj=108]: the master played Qf2; under μ our Rb1 plays the role of that game’s Qf3 (a **cross-type** correspondence); cross-type roles: our Bg2 as their Rc1, our Pa2 as their Bg2; transferred move b1c1. Synthesis: 2 retrieved analogs transferred to Bd5+ (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



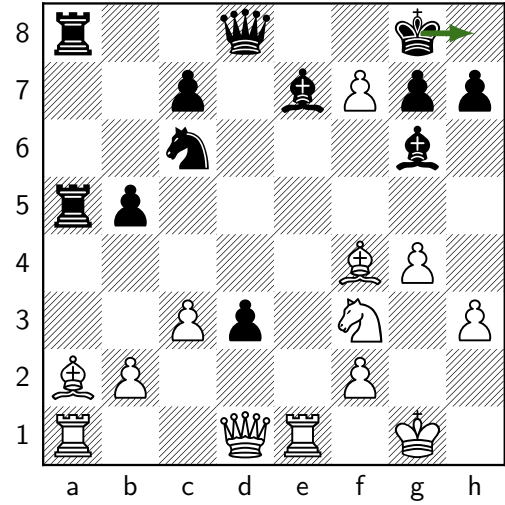
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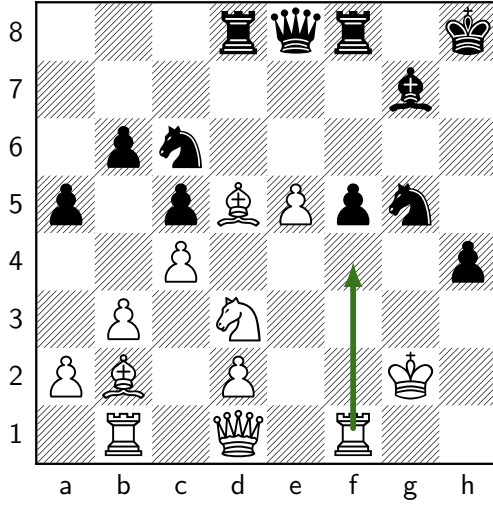


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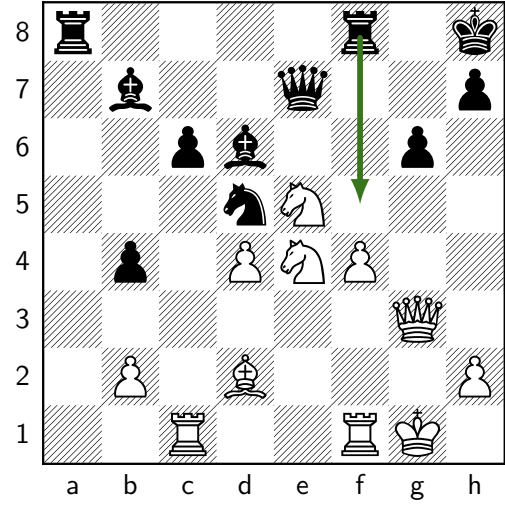


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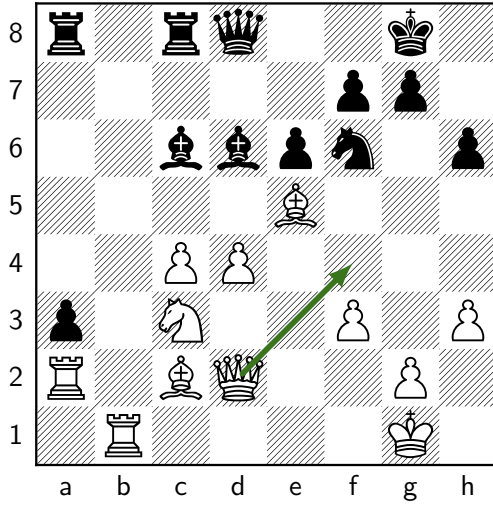
Figure 19: **Move 25, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2190` (ply 37) [structural distance $d=78$, QAP obj=45]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bb2* as their *Rc8*, our *ng5* as their *Pg7*; transferred move **g1g2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0003#2531` (ply 37) [structural distance $d=78$, QAP obj=45]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bb2* as their *Rc8*, our *ng5* as their *Pg7*; transferred move **g1g2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#3090` (ply 45) [structural distance $d=72$, QAP obj=12]: the master played *Kh8*; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Bb2* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move **g1g2 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to Kg2 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



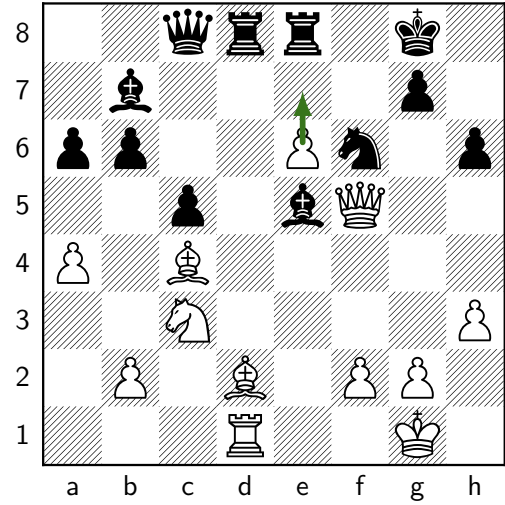
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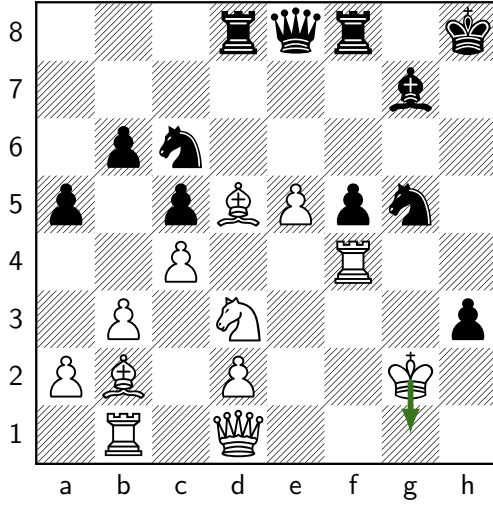


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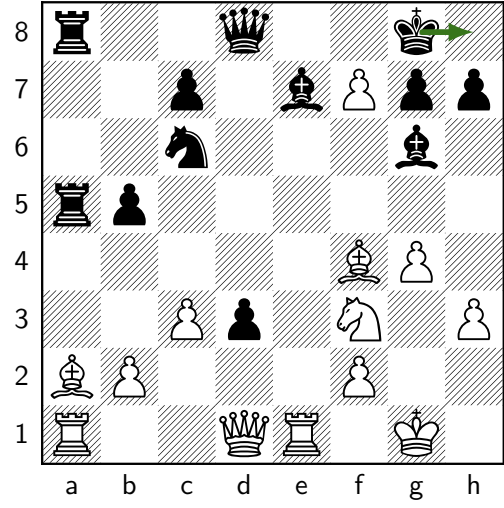


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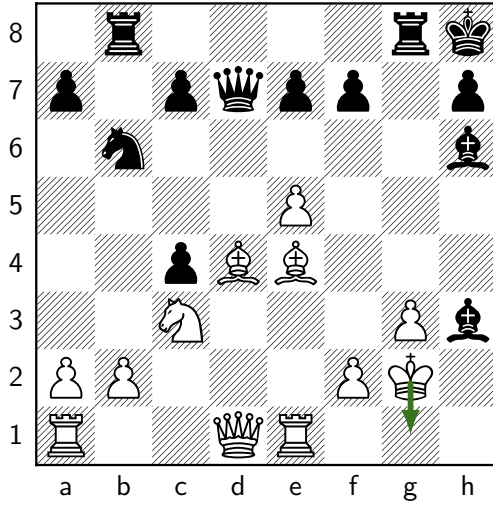
Figure 20: **Move 26, White: Rf4**. (a) Our position; Melanie plays Rf4 (f1f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3064` (ply 51) [structural distance $d=75$, QAP obj=66]: the master played Rf5; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Qd1 as their Ra8, our Nd3 as their Bb7; transferred move f1f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2553` (ply 50) [structural distance $d=78$, QAP obj=46]: the master played Qf4; under μ our Rf1 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our qe8 as their Rb1, our bg7 as their Ra2; transferred move f1f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1952` (ply 40) [structural distance $d=84$, QAP obj=78]: the master played e7+; under μ our Pe5 plays the role of that game’s Pe6; cross-type roles: our Nd3 as their Rd1, our Pe5 as their Bd2; transferred move e5e6. Synthesis: 2 retrieved analogs transferred to Rf4 (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



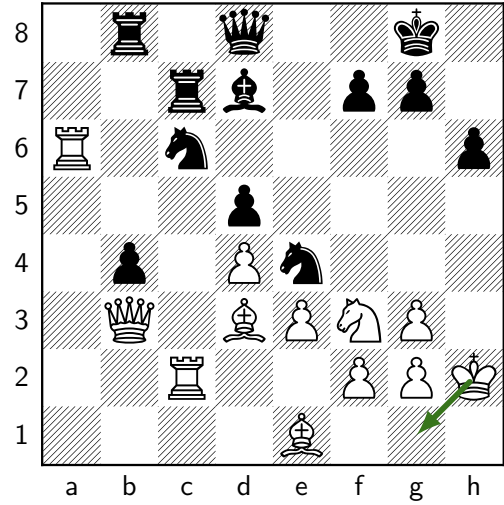
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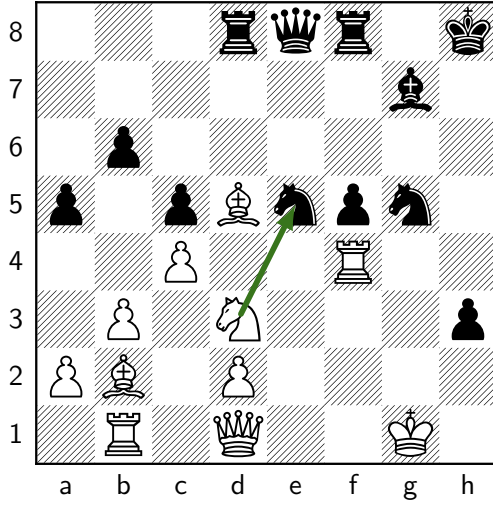


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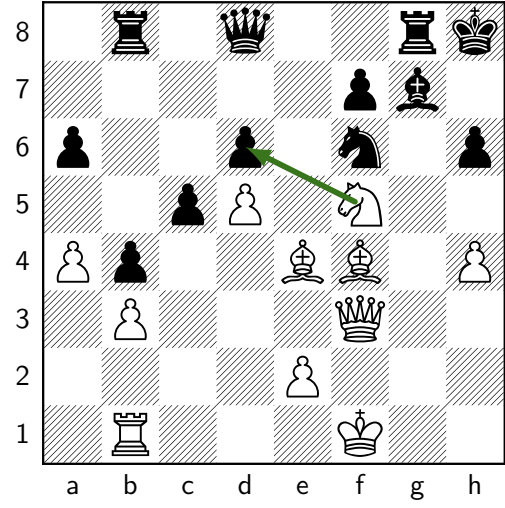


(d)

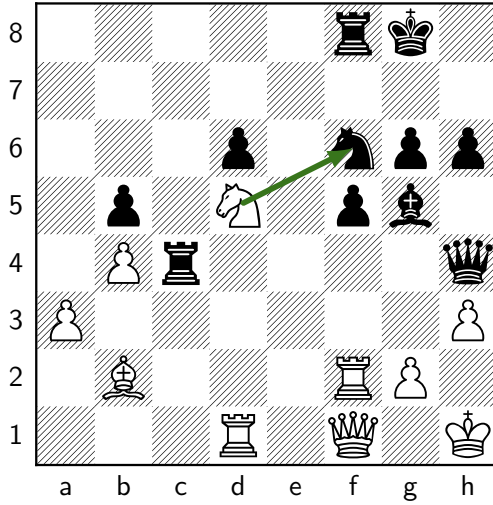
Figure 21: **Move 27, White: Kg1.** (a) Our position; Melanie plays Kg1 (**g2g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3090` (ply 45) [structural distance $d=93$, QAP obj=102]: the master played Kh8; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Bd5 as their Ra8, our Nd3 as their Qd8; transferred move **g2g1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#3853` (ply 40) [structural distance $d=94$, QAP obj=91]: the master played Kg1; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our rf8 as their Qd1, our pc5 as their Nc3; transferred move **g2g1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0001#2042` (ply 60) [structural distance $d=95$, QAP obj=34]: the master played Kg1; under μ our Kg2 plays the role of that game’s Kh2; cross-type roles: our bg7 as their Rc2, our Qd1 as their Pf2; transferred move **g2g1 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to Kg1 (summed QAP objective 226, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



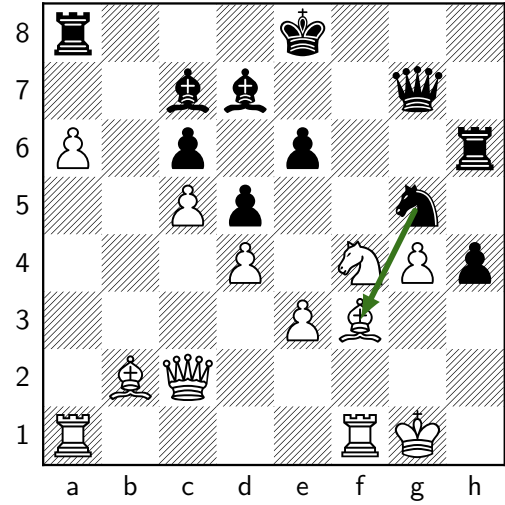
(a)



(b)

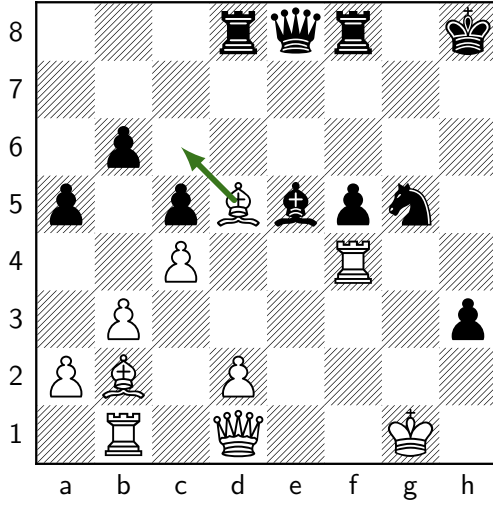


(c)

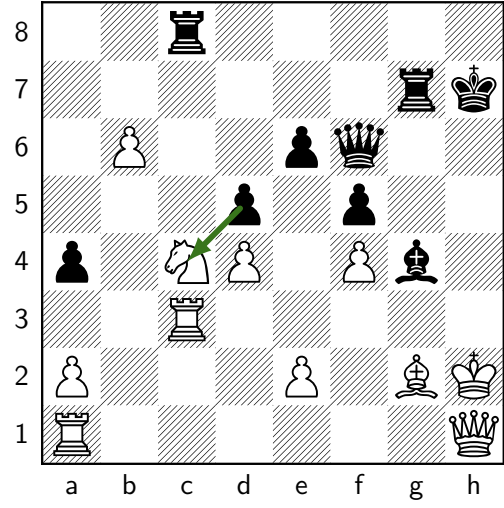


(d)

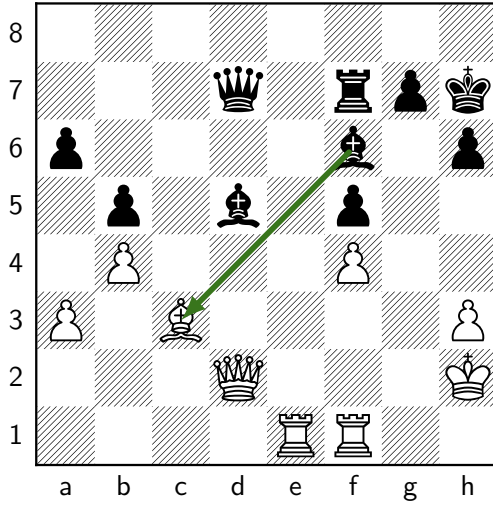
Figure 22: **Move 28, White: Nxe5.** (a) Our position; Melanie plays Nxe5 (d3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#748` (ply 58) [structural distance $d=104$, QAP obj=78]: the master played Nxd6; under μ our Nd3 plays the role of that game’s Nf5; cross-type roles: our Qd1 as their Rb1, our Bd5 as their Qf3; transferred move d3e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1845` (ply 62) [structural distance $d=105$, QAP obj=76]: the master played Nxf6+; under μ our Nd3 plays the role of that game’s Nd5; cross-type roles: our pf5 as their Rf2, our rf8 as their Pg2; transferred move d3e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#534` (ply 51) [structural distance $d=99$, QAP obj=70]: the master played Nxf3+; under μ our Nd3 plays the role of that game’s Ng4; cross-type roles: our Qd1 as their Bd7, our rd8 as their Qg7; transferred move d3e5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nxe5 (summed QAP objective 320, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



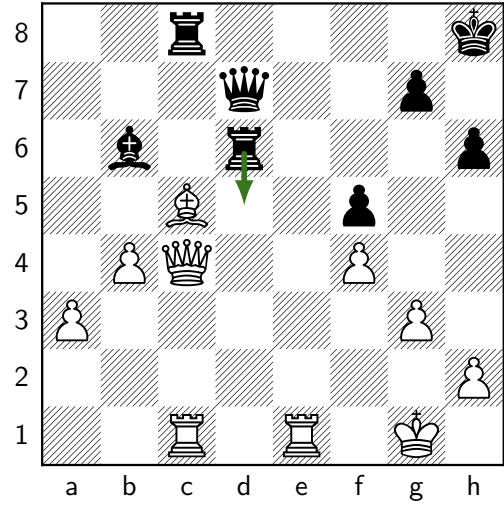
(a)



(b)

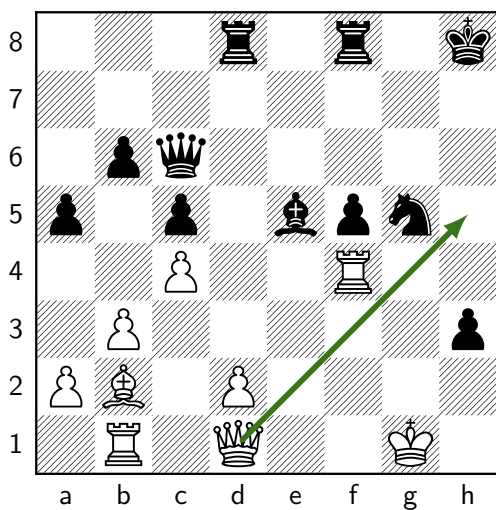


(c)

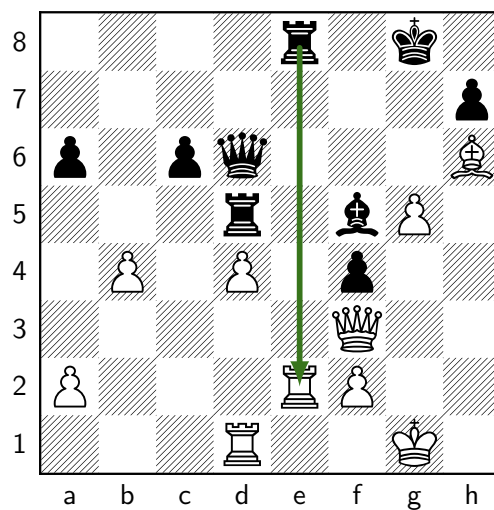


(d)

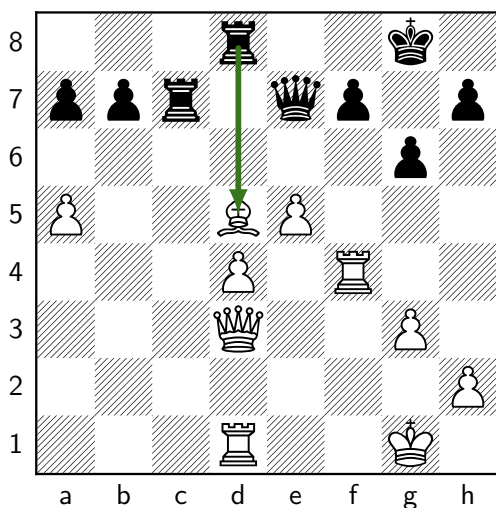
Figure 23: **Move 29, White: Bc6.** (a) Our position; Melanie plays Bc6 (d5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#604` (ply 73) [structural distance $d=84$, QAP obj=31]: the master played `dxc4`; under μ our `Bd5` plays the role of that game’s `Pd4` (a **cross-type** correspondence); cross-type roles: our `Bb2` as their `Rc8`, our `Qd1` as their `Rg7`; transferred move `d5c6` (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#109` (ply 61) [structural distance $d=89$, QAP obj=26]: the master played `Bxc3`; under μ our `Bd5` plays the role of that game’s `Bf3`; cross-type roles: our `rf8` as their `Qd7`, our `ng5` as their `Pg7`; transferred move `d5c6` (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3489` (ply 61) [structural distance $d=87$, QAP obj=74]: the master played `Rd5`; under μ our `Rf4` plays the role of that game’s `Rd3`; cross-type roles: our `Bb2` as their `Rc8`, our `Rb1` as their `Qd7`; transferred move `f4f5`. Synthesis: 2 retrieved analogs transferred to Bc6 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



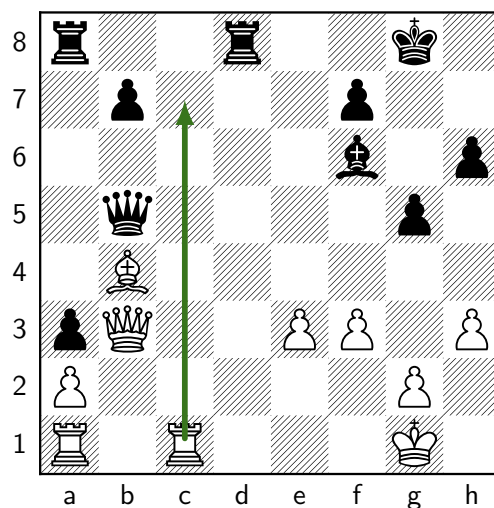
(a)



(b)

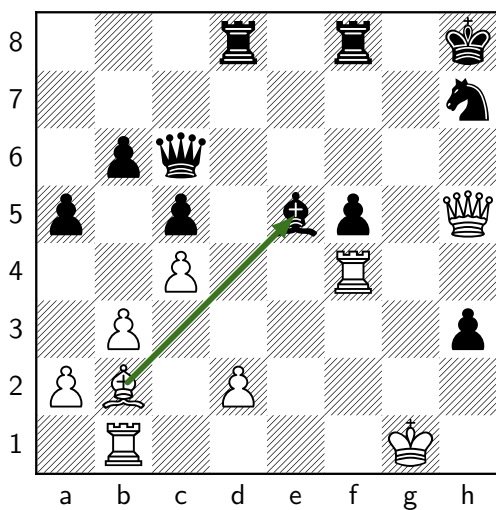


(c)

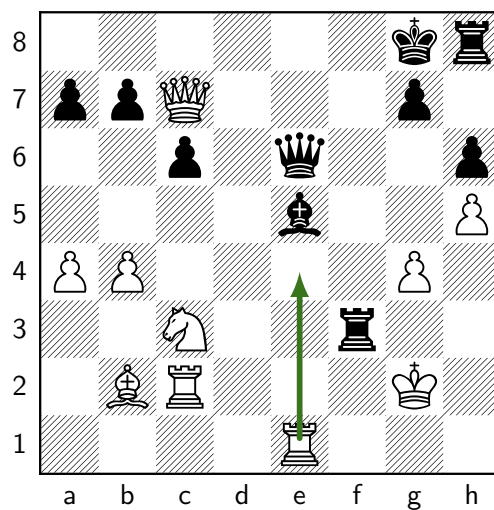


(d)

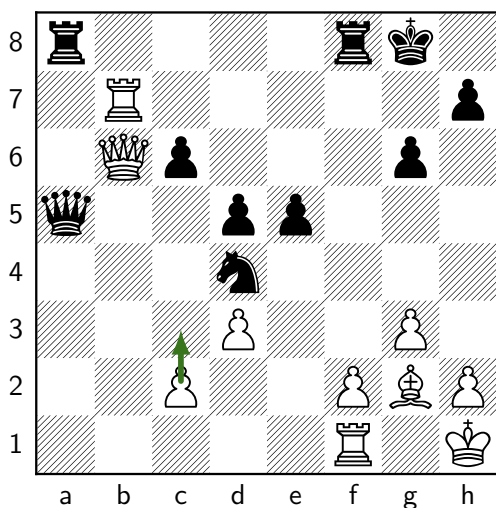
Figure 24: **Move 30, White: Qh5+.** (a) Our position; Melanie plays Qh5+ (d1h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2606` (ply 65) [structural distance $d=71$, QAP obj=47]: the master played Rxe2; under μ our Qd1 plays the role of that game's Re1 (a **cross-type** correspondence); cross-type roles: our Pc4 as their Qd6, our qc6 as their bh6; transferred move d1h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#184` (ply 53) [structural distance $d=66$, QAP obj=16]: the master played Rxd5; under μ our Qd1 plays the role of that game's Rd1 (a **cross-type** correspondence); cross-type roles: our ng5 as their Rc7, our rf8 as their Qe7; transferred move d1h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#422` (ply 60) [structural distance $d=64$, QAP obj=15]: the master played Rc7; under μ our Qd1 plays the role of that game's Rc1 (a **cross-type** correspondence); cross-type roles: our Pc4 as their Ra1, our Bb2 as their Qb3; transferred move d1h5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qh5+ (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



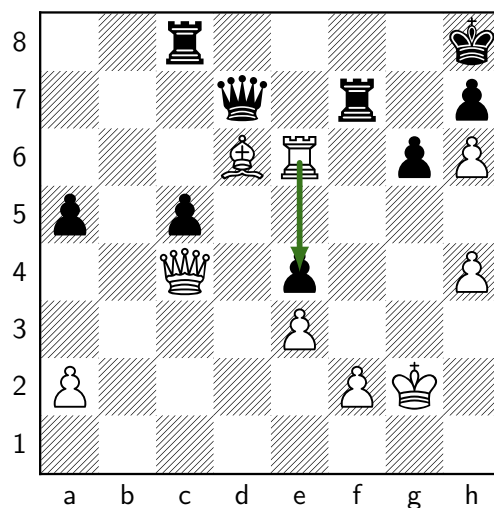
(a)



(b)

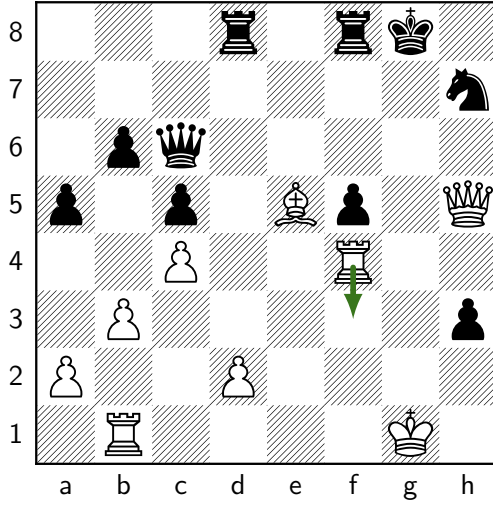


(c)

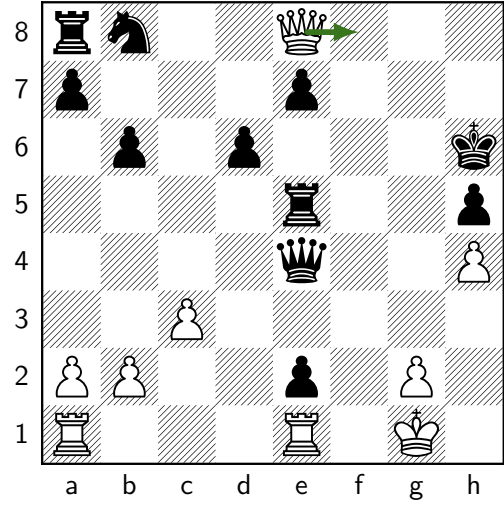


(d)

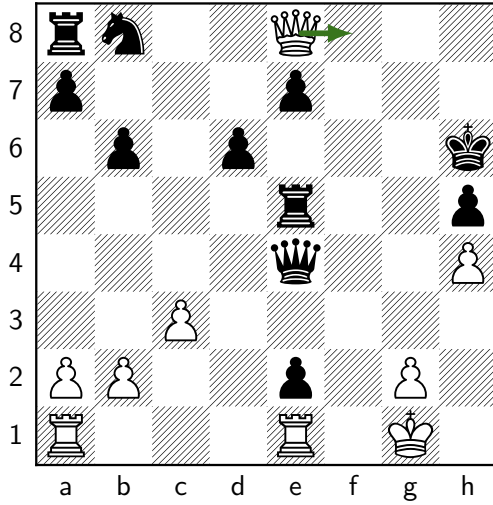
Figure 25: **Move 31, White: Bxe5+.** (a) Our position; Melanie plays Bxe5+ (b2e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2682` (ply 62) [structural distance $d=90$, QAP obj=46]: the master played **Re4**; under μ our *Bb2* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Pd2* as their *Bb2*, our *Rf4* as their *Nc3*; transferred move **b2e5 (endorses this move)**. (c) Retrieved from `ccrl_top_p0003#123` (ply 50) [structural distance $d=90$, QAP obj=13]: the master played **c3**; under μ our *Bb2* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *pa5* as their *Rf1*, our *qc6* as their *Bg2*; transferred move **b2e5 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#3914` (ply 66) [structural distance $d=83$, QAP obj=58]: the master played **Rxe4**; under μ our *Rf4* plays the role of that game’s *Re6*; cross-type roles: our *Rb1* as their *Pa2*, our *rd8* as their *Pe3*; transferred move **f4h4**. Synthesis: 2 retrieved analogs transferred to Bxe5+ (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



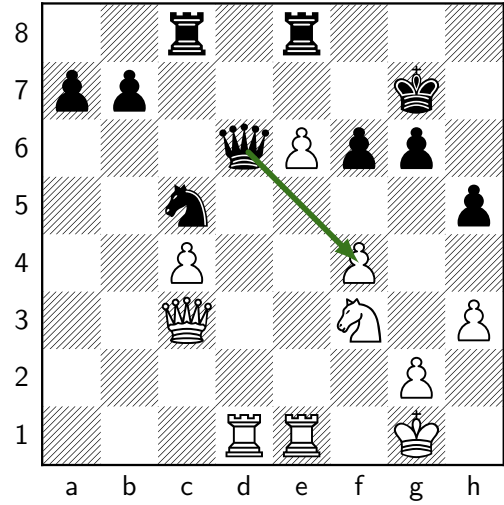
(a)



(b)

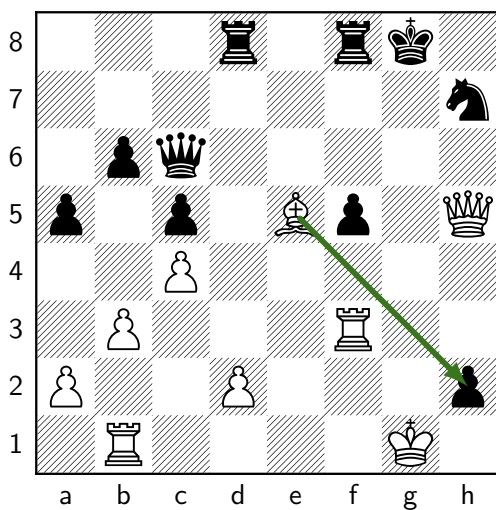


(c)

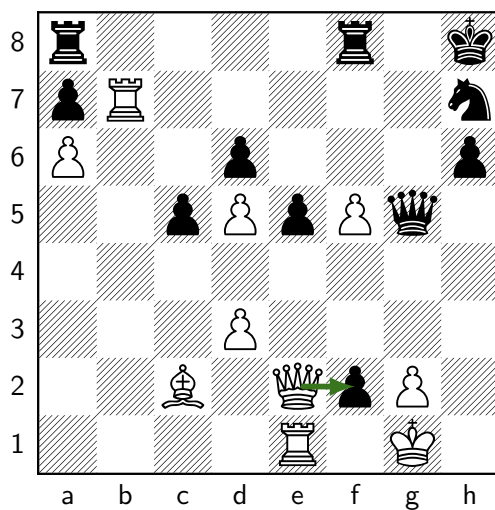


(d)

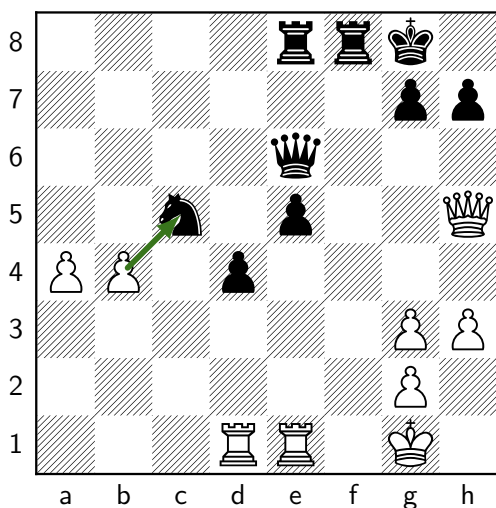
Figure 26: **Move 32, White: Rf3.** (a) Our position; Melanie plays Rf3 (f4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2676` (ply 52) [structural distance $d=62$, QAP obj=96]: the master played Qf8+; under μ our Rf4 plays the role of that game’s Qe8 (a **cross-type** correspondence); cross-type roles: our Pc4 as their Ra1, our Rb1 as their Pa2; transferred move f4h4. (c) Retrieved from `ccr1_top_p0002#2695` (ply 52) [structural distance $d=62$, QAP obj=96]: the master played Qf8+; under μ our Rf4 plays the role of that game’s Qe8 (a **cross-type** correspondence); cross-type roles: our Pc4 as their Ra1, our Rb1 as their Pa2; transferred move f4h4. (d) Retrieved from `ccr1_top_p0003#187` (ply 69) [structural distance $d=69$, QAP obj=64]: the master played Qxf4; under μ our Qh5 plays the role of that game’s Qd3; cross-type roles: our Be5 as their Rc8, our Rb1 as their Pa7; transferred move h5h3. Synthesis: no retrieved analog transferred to Rf3 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



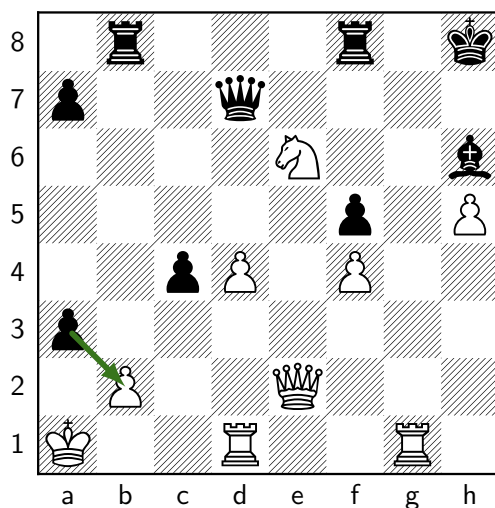
(a)



(b)

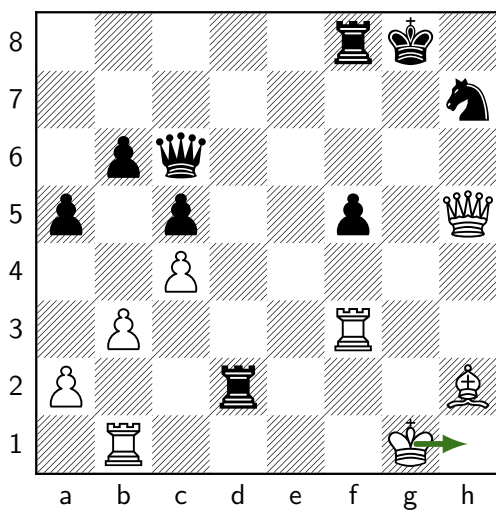


(c)

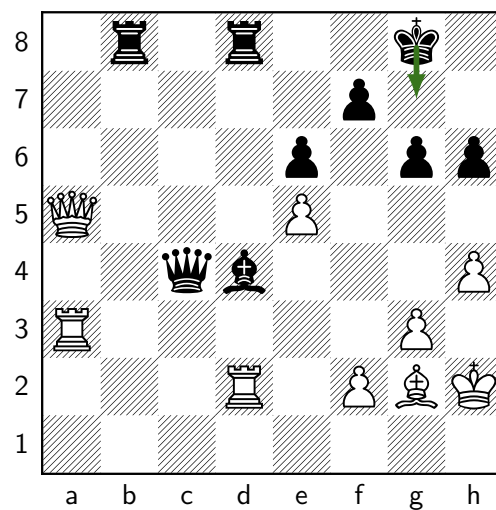


(d)

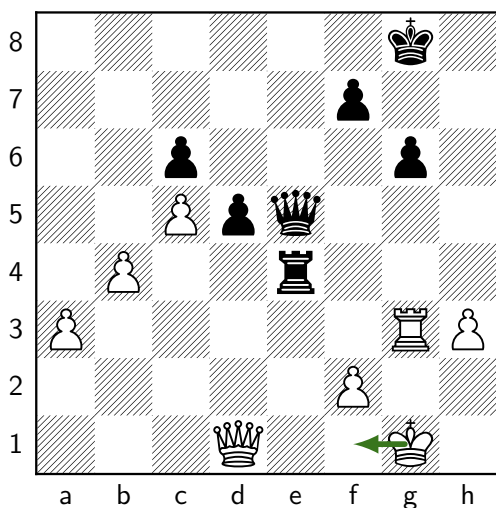
Figure 27: **Move 33, White: Bxh2.** (a) Our position; Melanie plays Bxh2 (e5h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#999` (ply 64) [structural distance $d=72$, QAP obj=83]: the master played Qxf2; under μ our *Be5* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *pb6* as their *Bc2*, our *qc6* as their *pc5*; transferred move e5h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1614` (ply 62) [structural distance $d=70$, QAP obj=26]: the master played bxc5; under μ our *Be5* plays the role of that game’s *Pb4* (a **cross-type** correspondence); cross-type roles: our *Qh5* as their *Rd1*, our *Rb1* as their *Pa4*; transferred move e5h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1486` (ply 65) [structural distance $d=71$, QAP obj=19]: the master played axb2+; under μ our *Be5* plays the role of that game’s *Pa6* (a **cross-type** correspondence); cross-type roles: our *Qh5* as their *Rb8*, our *Rf3* as their *Qd7*; transferred move e5h2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bxh2 (summed QAP objective 151, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



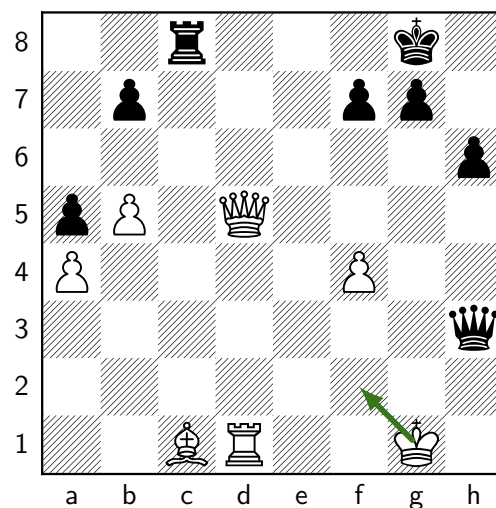
(a)



(b)

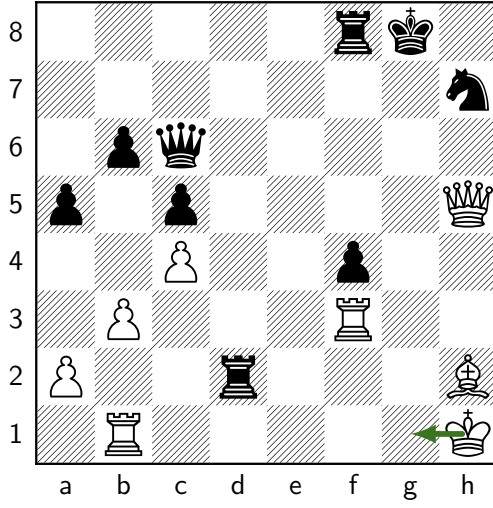


(c)

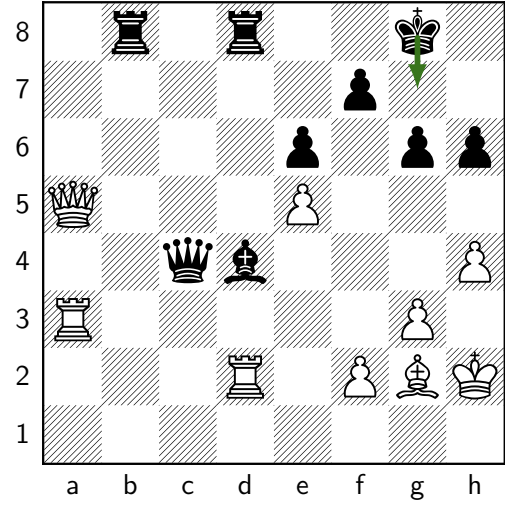


(d)

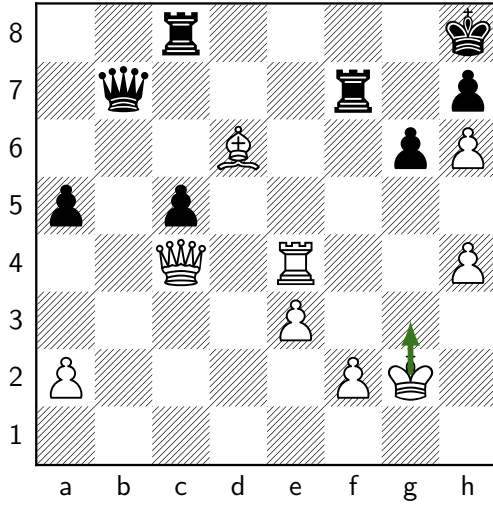
Figure 28: **Move 34, White: Kh1**. (a) Our position; Melanie plays Kh1 (**g1h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#691` (ply 69) [structural distance $d=57$, QAP obj=63]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our nh7 as their Rd8, our Rb1 as their Pg6; transferred move g1h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3287` (ply 74) [structural distance $d=56$, QAP obj=54]: the master played Kf1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Bh2 as their Qd1, our Qh5 as their Pf2; transferred move g1h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#416` (ply 58) [structural distance $d=56$, QAP obj=35]: the master played Kf2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Qh5 as their Rd1, our rf8 as their gh3; transferred move g1h1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kh1 (summed QAP objective 233, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



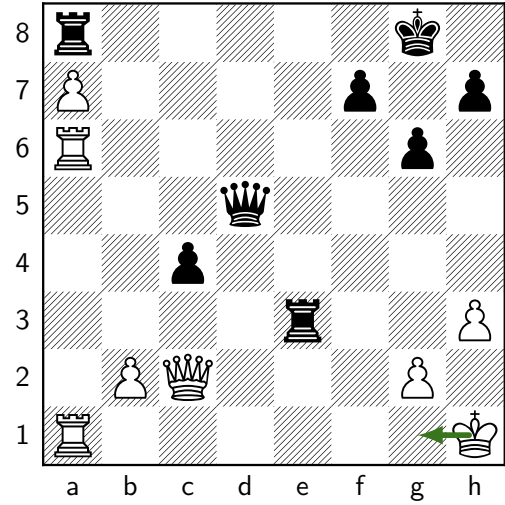
(a)



(b)

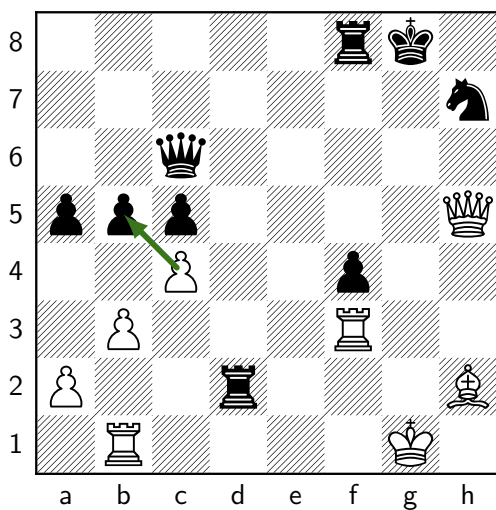


(c)

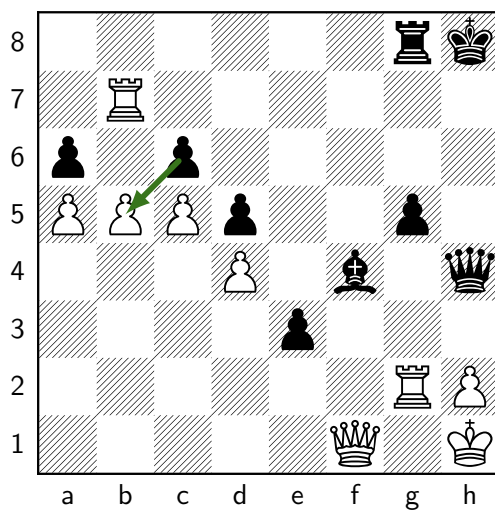


(d)

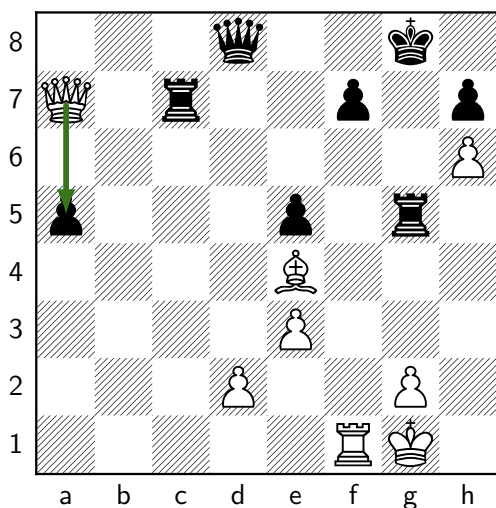
Figure 29: **Move 35, White: Kg1.** (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#691` (ply 69) [structural distance $d=59$, QAP obj=109]: the master played Kg7; under μ our *Kh1* plays the role of that game’s *Kg1*; cross-type roles: our *pb6* as their *Rb8*, our *nh7* as their *Rd8*; transferred move **h1g1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#3914` (ply 68) [structural distance $d=55$, QAP obj=87]: the master played Kg3; under μ our *Kh1* plays the role of that game’s *Kg2*; cross-type roles: our *rf8* as their *Pe3*, our *Rf3* as their *pc5*; transferred move **h1g1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#3879` (ply 70) [structural distance $d=54$, QAP obj=34]: the master played Kg1; under μ our *Kh1* plays the role of that game’s *Kh1*; cross-type roles: our *rd2* as their *Qc2*, our *Pc4* as their *re3*; transferred move **h1g1 (endorses this move)**. Synthesis: 8 retrieved analogs transferred to Kg1 (summed QAP objective 332, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



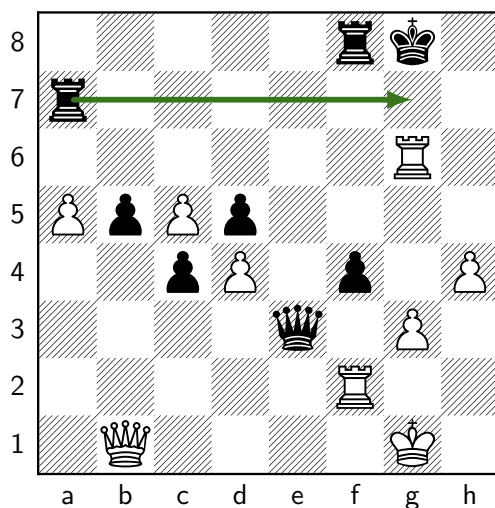
(a)



(b)



(c)



(d)

Figure 30: **Move 36, White: cxb5**. (a) Our position; Melanie plays cxb5 (c4b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3413 (ply 75) [structural distance $d=67$, QAP obj=87]: the master played cxb5; under μ our Pc_4 plays the role of that game’s Pc_3 ; cross-type roles: our Qh_5 as their Rg_8 , our qc_6 as their rb_7 ; transferred move c4b5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3664 (ply 56) [structural distance $d=62$, QAP obj=20]: the master played Qxa5; under μ our Pc_4 plays the role of that game’s Qa_7 (a **cross-type** correspondence); cross-type roles: our Bh_2 as their Rf_1 , our rf_8 as their Be_4 ; transferred move c4b5 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#163 (ply 85) [structural distance $d=59$, QAP obj=18]: the master played Rg7; under μ our Pc_4 plays the role of that game’s Ra_2 (a **cross-type** correspondence); cross-type roles: our Qh_5 as their rg_6 , our nh_7 as their Pb_5 ; transferred move c4b5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to cxb5 (summed QAP objective 186, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

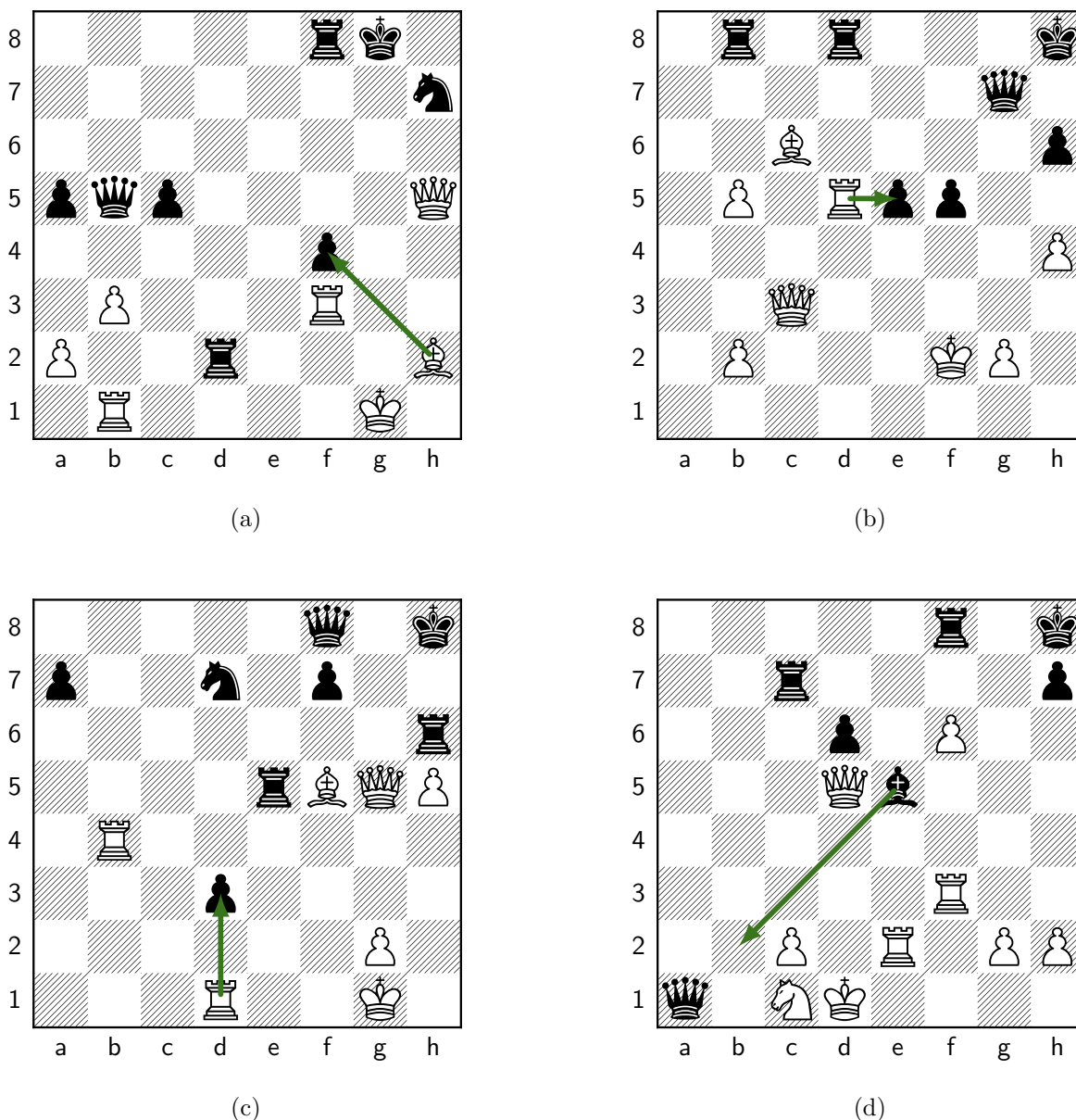


Figure 31: **Move 37, White: Bxf4.** (a) Our position; Melanie plays Bxf4 (h2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#758` (ply 64) [structural distance $d=64$, QAP obj=39]: the master played Rxe5; under μ our Bh2 plays the role of that game’s Rd5 (a **cross-type** correspondence); cross-type roles: our Rb1 as their Pb2, our nh7 as their Pg2; transferred move h2f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#830` (ply 86) [structural distance $d=64$, QAP obj=17]: the master played Rxd3; under μ our Bh2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Rf3 as their Pg2, our Pb3 as their Rb4; transferred move h2f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2303` (ply 73) [structural distance $d=46$, QAP obj=64]: the master played Bb2; under μ our Bh2 plays the role of that game’s Be4; cross-type roles: our qb5 as their Rc7, our rd2 as their qd5; transferred move h2g3. Synthesis: 2 retrieved analogs transferred to Bxf4 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

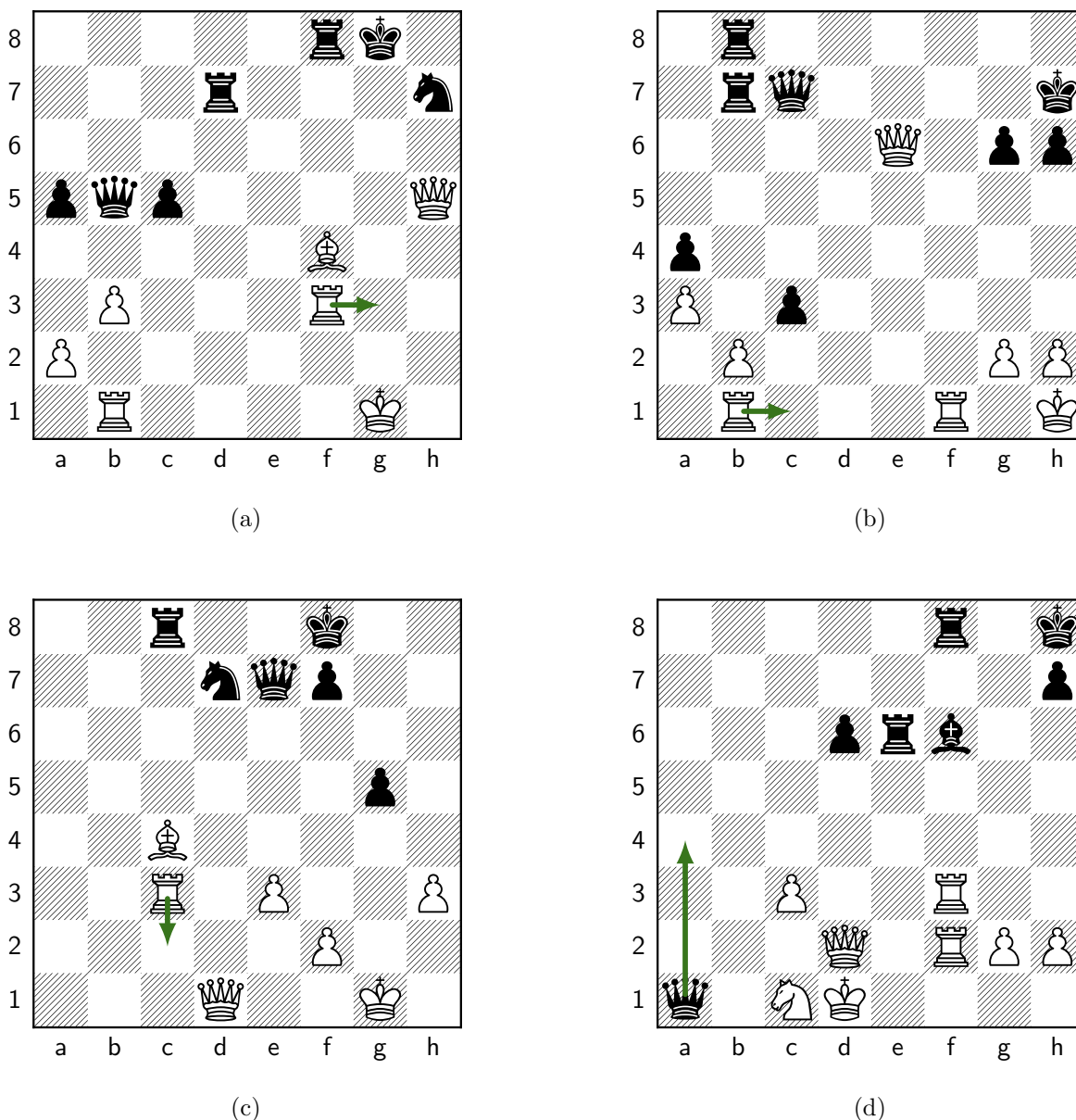
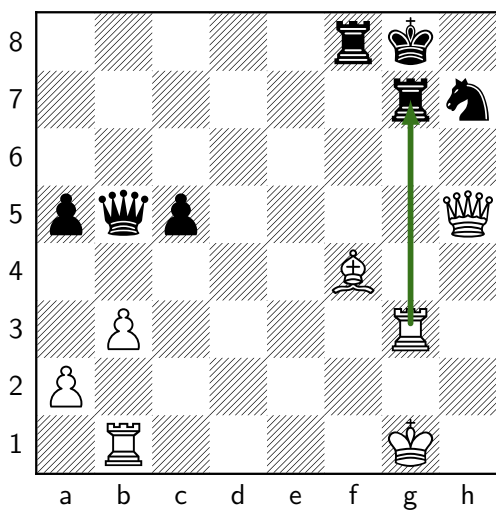
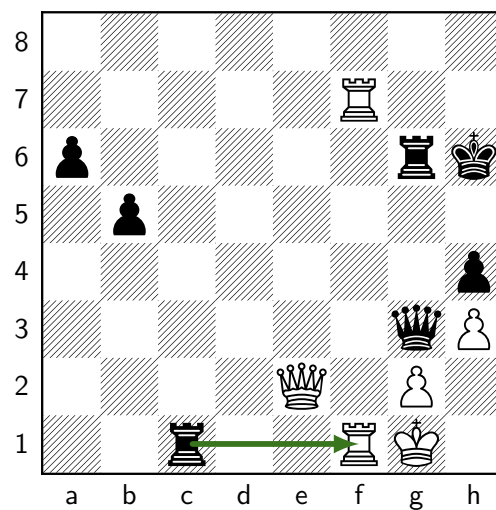


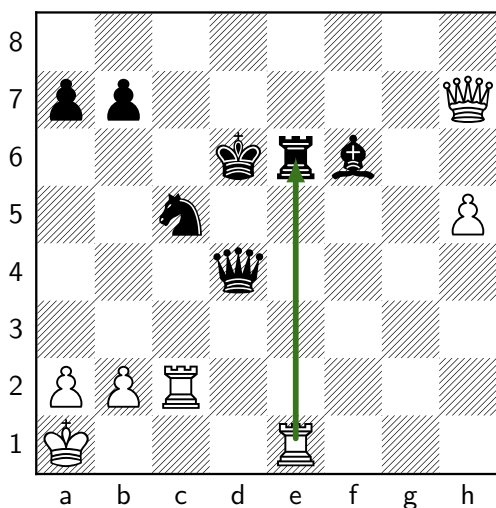
Figure 32: **Move 38, White: Rg3+.** (a) Our position; Melanie plays Rg3+ (f3g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1609` (ply 70) [structural distance $d=69$, QAP obj=21]: the master played Rbc1; under μ our *Rf3* plays the role of that game’s *Rb1*; cross-type roles: our *Bf4* as their *Pb2*, our *Rb1* as their *Pg2*; transferred move f3g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1022` (ply 80) [structural distance $d=67$, QAP obj=46]: the master played Rc2; under μ our *Rf3* plays the role of that game’s *Rc3*; cross-type roles: our *Rb1* as their *Ph3*, our *rd7* as their *pf7*; transferred move f3h3. (d) Retrieved from `ccr1_top_p0000#2303` (ply 87) [structural distance $d=64$, QAP obj=44]: the master played Qa4+; under μ our *Qh5* plays the role of that game’s *Qa8*; cross-type roles: our *pc5* as their *Rf8*, our *Rb1* as their *Ph7*; transferred move h5h4. Synthesis: 1 retrieved analog transferred to Rg3+ (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



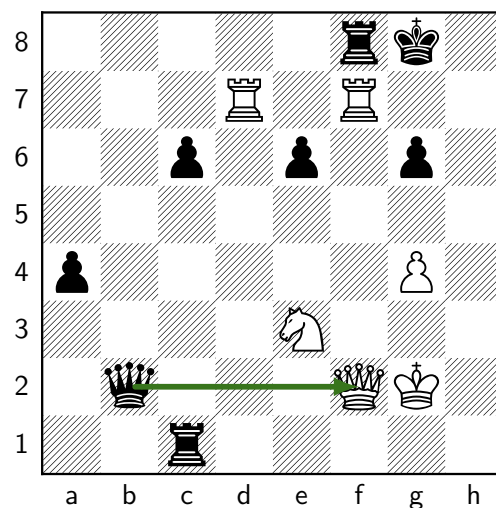
(a)



(b)

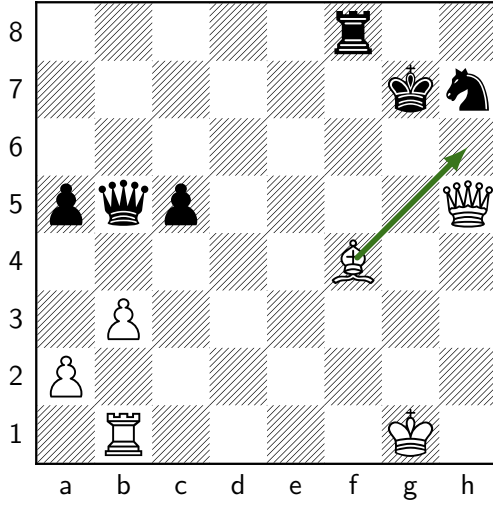


(c)

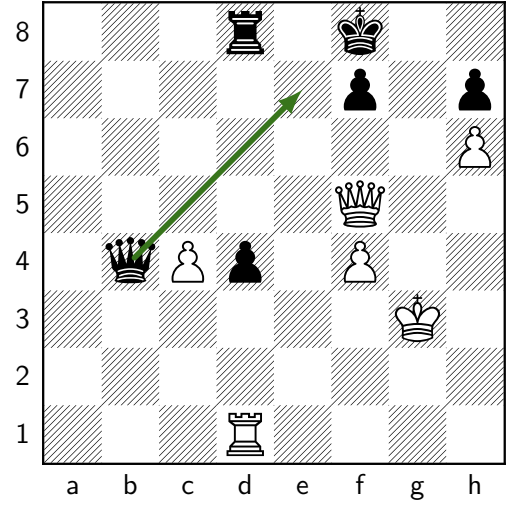


(d)

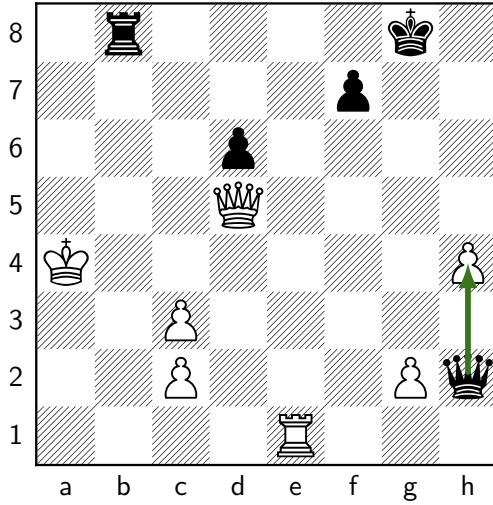
Figure 33: **Move 39, White: R_xg7+.** (a) Our position; Melanie plays R_xg7+ (g3g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2515` (ply 69) [structural distance $d=75$, QAP obj=74]: the master played R_xf1+; under μ our R_g3 plays the role of that game's R_c8; cross-type roles: our n_h7 as their r_f7, our Q_h5 as their R_g6; transferred move g3g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#541` (ply 72) [structural distance $d=79$, QAP obj=62]: the master played R_xe6+; under μ our R_g3 plays the role of that game's R_e1; cross-type roles: our Q_h5 as their R_c2, our R_b1 as their P_h5; transferred move g3g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2821` (ply 75) [structural distance $d=74$, QAP obj=28]: the master played Q_xf2+; under μ our R_g3 plays the role of that game's Q_b7 (a **cross-type** correspondence); cross-type roles: our Q_h5 as their R_f8, our q_b5 as their r_d7; transferred move g3g7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to R_xg7+ (summed QAP objective 186, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



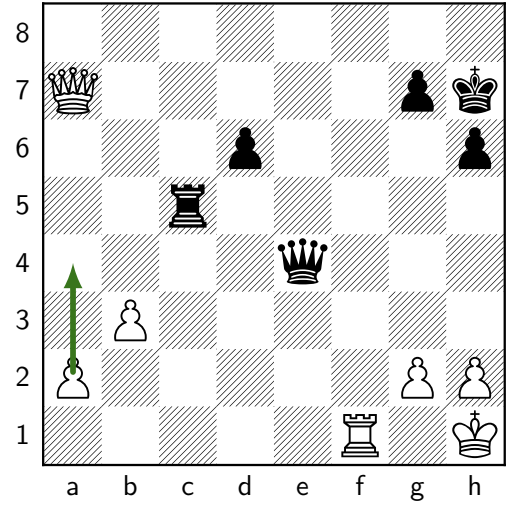
(a)



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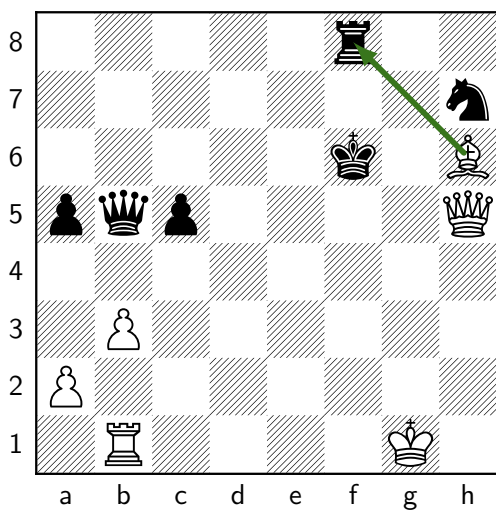


(c)

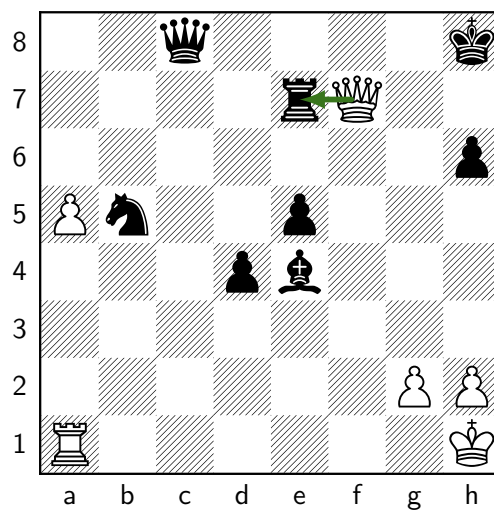


(d)

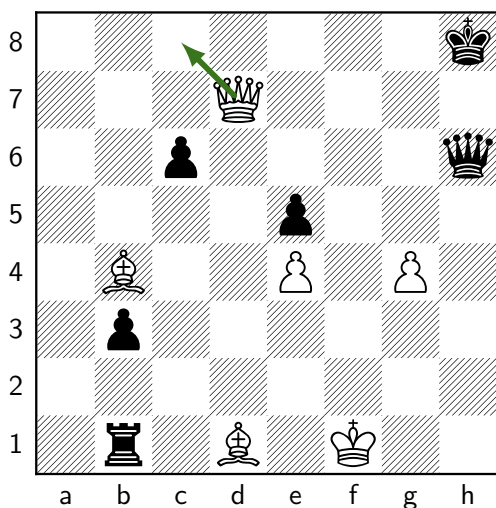
Figure 34: **Move 40, White: Bh6+.** (a) Our position; Melanie plays Bh6+ (f4h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1749` (ply 93) [structural distance $d=44$, QAP obj=53]: the master played Qe7; under μ our Qh5 plays the role of that game’s Qb5; cross-type roles: our nh7 as their pc4, our qb5 as their rd1; transferred move h5h1. (c) Retrieved from `ccr1_top_p0001#3950` (ply 95) [structural distance $d=47$, QAP obj=44]: the master played Qxh4+; under μ our Qh5 plays the role of that game’s Qh7; cross-type roles: our nh7 as their ph4, our rf8 as their pc2; transferred move h5h4. (d) Retrieved from `ccr1_top_p0001#994` (ply 74) [structural distance $d=49$, QAP obj=44]: the master played a4; under μ our Bf4 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our nh7 as their rc5, our rf8 as their pd6; transferred move f4g5. Synthesis: no retrieved analog transferred to Bh6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



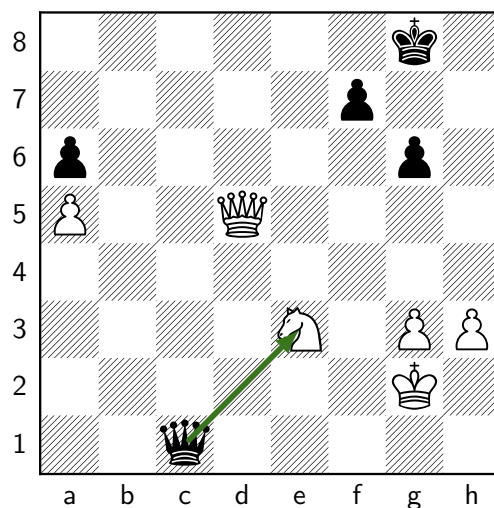
(a)



(b)



(c)



(d)

Figure 35: **Move 41, White: Bxf8**. (a) Our position; Melanie plays Bxf8 (h6f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1849` (ply 80) [structural distance $d=54$, QAP obj=27]: the master played Qxe7; under μ our Bh6 plays the role of that game’s Qf7 (a **cross-type** correspondence); cross-type roles: our qb5 as their be4, our Qh5 as their nb5; transferred move h6f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1493` (ply 90) [structural distance $d=54$, QAP obj=20]: the master played Qc8+; under μ our Bh6 plays the role of that game’s Qd7 (a **cross-type** correspondence); cross-type roles: our qb5 as their rb1, our Pb3 as their Bd1; transferred move h6f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3175` (ply 85) [structural distance $d=51$, QAP obj=35]: the master played Qxe3; under μ our Bh6 plays the role of that game’s Qc8 (a **cross-type** correspondence); cross-type roles: our Rb1 as their Pa6, our rf8 as their ne3; transferred move h6f4. Synthesis: 4 retrieved analogs transferred to Bxf8 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

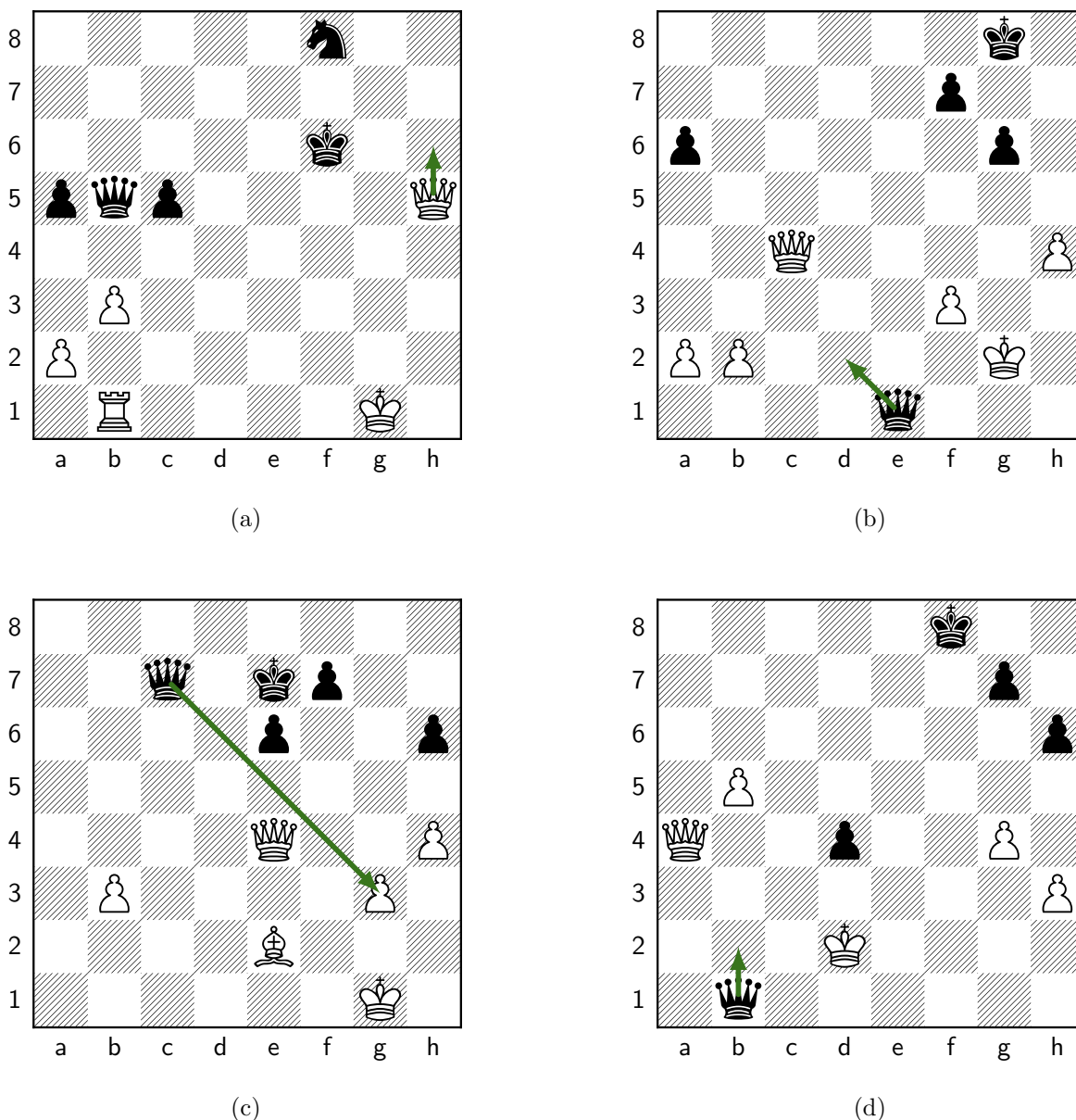
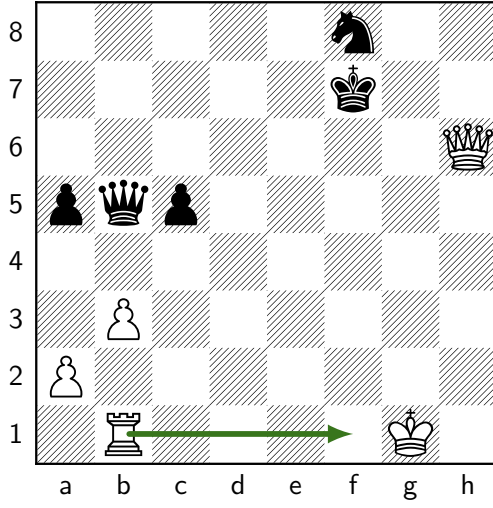
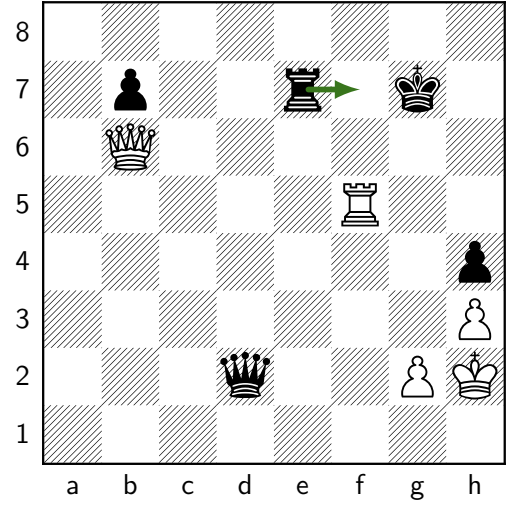


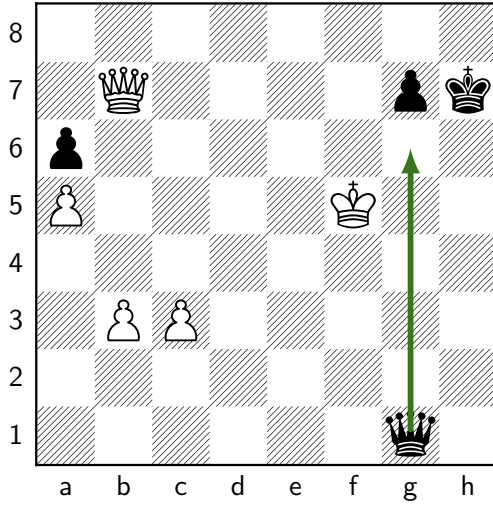
Figure 36: **Move 42, White: Qh6+.** (a) Our position; Melanie plays **Qh6+** (**h5h6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3599` (ply 71) [structural distance $d=33$, QAP obj=53]: the master played **Qd2+**; under μ our **Qh5** plays the role of that game's **Qe8**; cross-type roles: our **Rb1** as their **Pg6**, our **nf8** as their **pf3**; transferred move **h5h6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#606` (ply 73) [structural distance $d=31$, QAP obj=49]: the master played **Qxg3+**; under μ our **Qh5** plays the role of that game's **Qc2**; cross-type roles: our **Rb1** as their **Ph6**, our **nf8** as their **pb3**; transferred move **h5h6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3610` (ply 87) [structural distance $d=32$, QAP obj=23]: the master played **Qb2+**; under μ our **Qh5** plays the role of that game's **Qb8**; cross-type roles: our **Rb1** as their **Pg7**, our **nf8** as their **pg4**; transferred move **h5h6** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Qh6+** (summed QAP objective 147, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



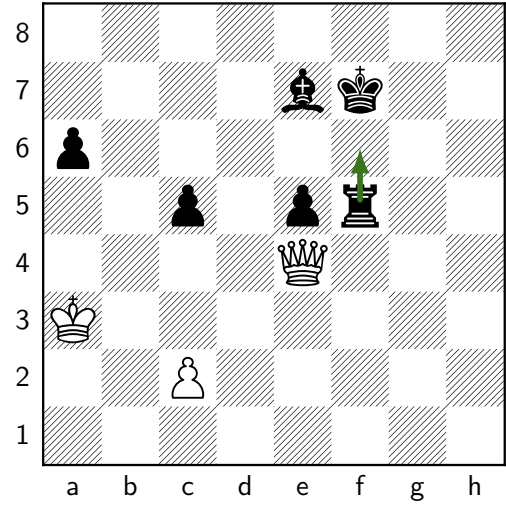
(a)



(b)

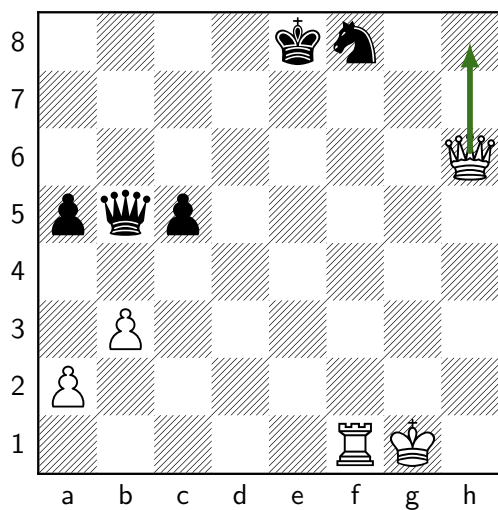


(c)

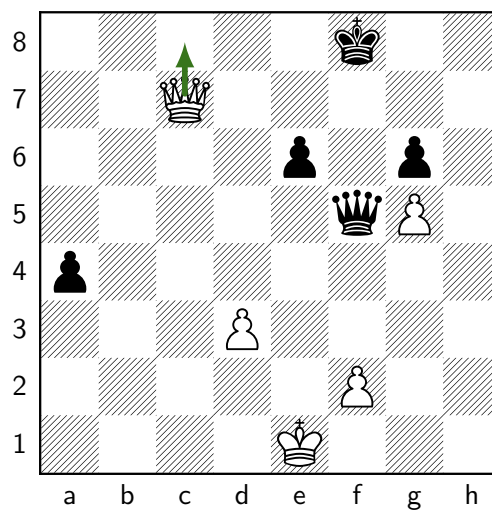


(d)

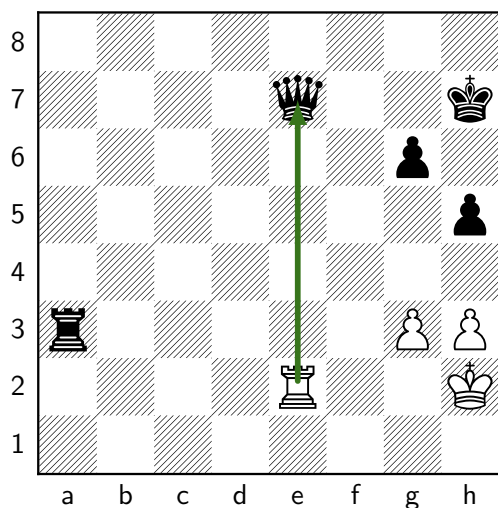
Figure 37: **Move 43, White: Rf1+.** (a) Our position; Melanie plays Rf1+ (b1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#51` (ply 101) [structural distance $d=29$, QAP obj=31]: the master played Rf7; under μ our Rb1 plays the role of that game’s Re2; cross-type roles: our nf8 as their Pb7, our Pa2 as their rf5; transferred move b1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#88` (ply 93) [structural distance $d=34$, QAP obj=14]: the master played Qg6+; under μ our Rb1 plays the role of that game’s Qg8 (a **cross-type** correspondence); transferred move b1f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2590` (ply 85) [structural distance $d=31$, QAP obj=14]: the master played Rf6; under μ our Rb1 plays the role of that game’s Rf4; cross-type roles: our Pa2 as their Be7, our Qh6 as their Pc5; transferred move b1f1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf1+ (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



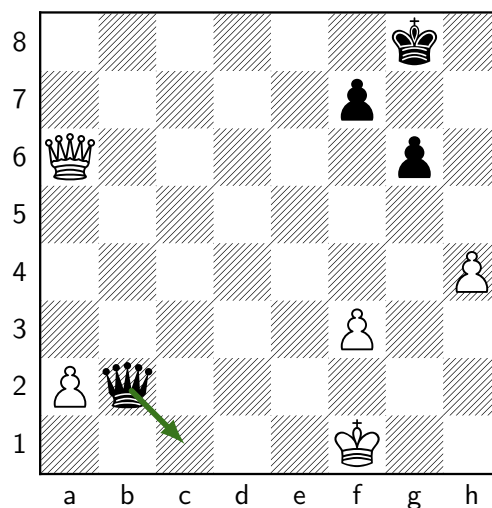
(a)



(b)

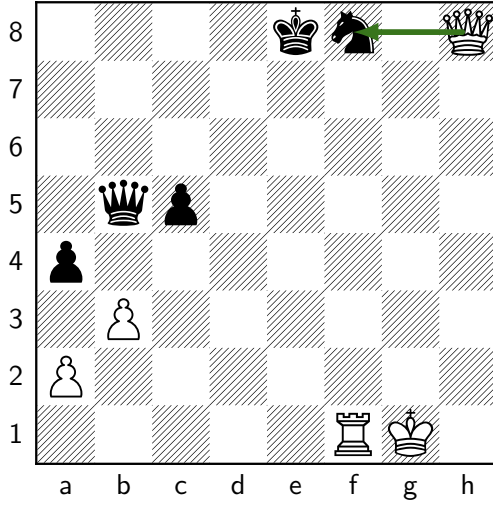


(c)

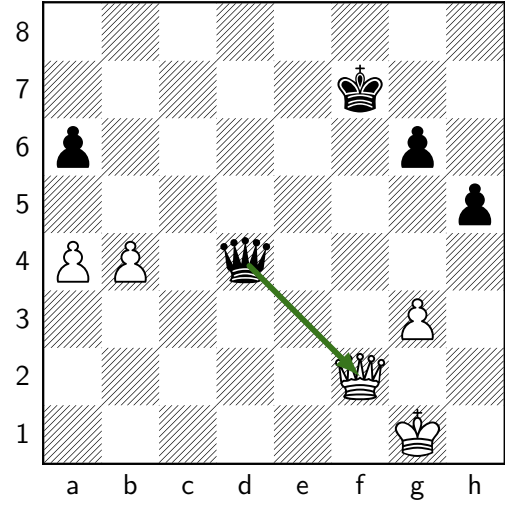


(d)

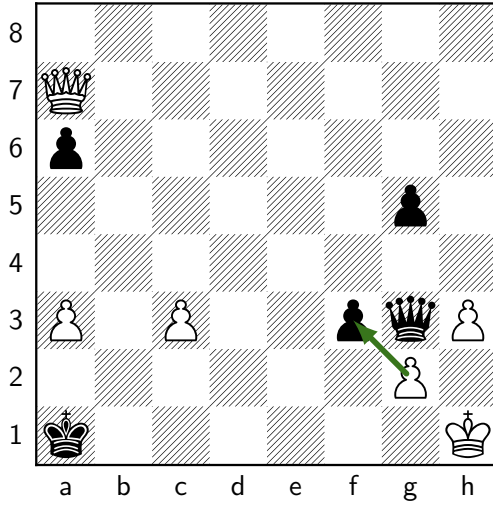
Figure 38: **Move 44, White: Qh8.** (a) Our position; Melanie plays **Qh8** (**h6h8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3604` (ply 102) [structural distance $d=37$, QAP obj=22]: the master played **Qc8+**; under μ our **Qh6** plays the role of that game’s **Qc7**; cross-type roles: our **nf8** as their **pa4**, our **Rf1** as their **Pg5**; transferred move **h6h8** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3965` (ply 102) [structural distance $d=32$, QAP obj=17]: the master played **Rxe7+**; under μ our **Qh6** plays the role of that game’s **Re2** (a **cross-type** correspondence); cross-type roles: our **qb5** as their **ra3**, our **nf8** as their **qe7**; transferred move **h6h8** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3599` (ply 75) [structural distance $d=34$, QAP obj=14]: the master played **Qc1+**; under μ our **Qh6** plays the role of that game’s **Qb7**; cross-type roles: our **Rf1** as their **pf3**, our **nf8** as their **pa2**; transferred move **h6h8** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to **Qh8** (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



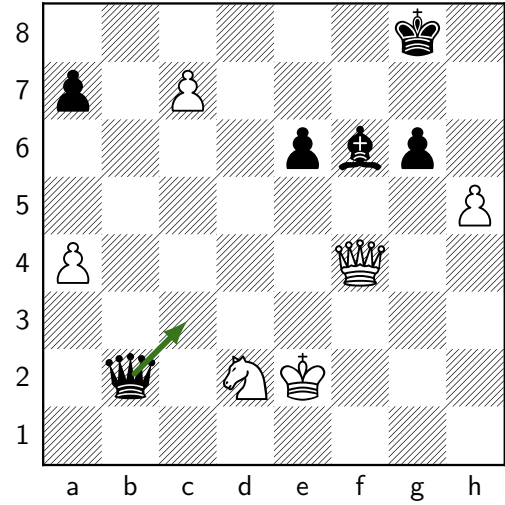
(a)



(b)

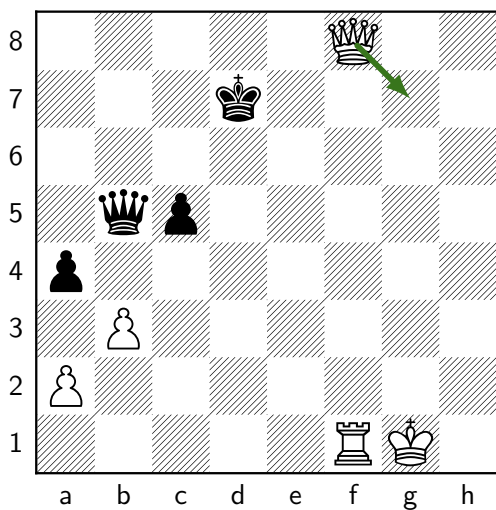


(c)

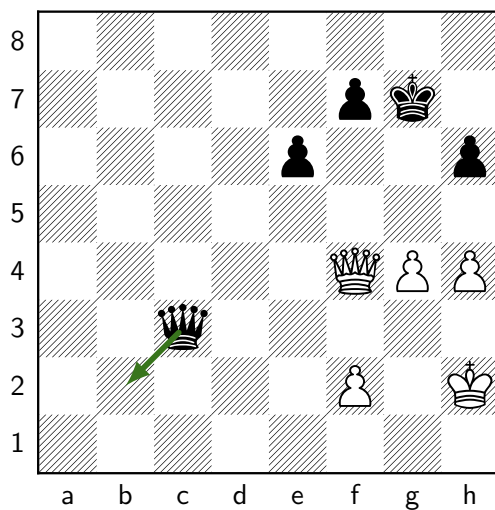


(d)

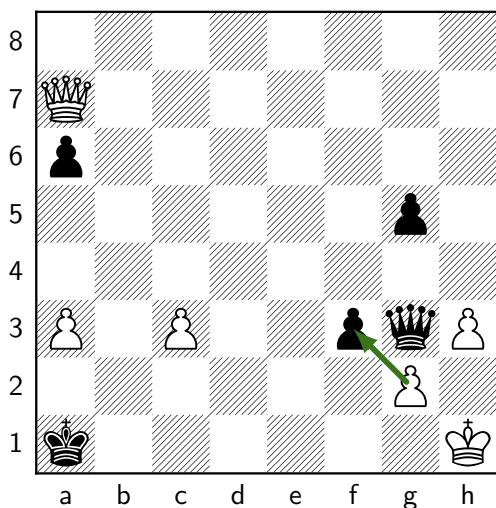
Figure 39: **Move 45, White: Qxf8+.** (a) Our position; Melanie plays Qxf8+ (h8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1524` (ply 87) [structural distance $d=45$, QAP obj=10]: the master played Qxf2+; under μ our Qh8 plays the role of that game’s Qd5; cross-type roles: our Rf1 as their Pg6, our qb5 as their pa4; transferred move h8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2798` (ply 80) [structural distance $d=35$, QAP obj=83]: the master played gxf3; under μ our Pb3 plays the role of that game’s Pg2; cross-type roles: our Rf1 as their Ph3, our nf8 as their pa6; transferred move b3a4. (d) Retrieved from `ccr1_top_p0000#1777` (ply 99) [structural distance $d=45$, QAP obj=81]: the master played Qc3; under μ our Qh8 plays the role of that game’s Qb7; cross-type roles: our Rf1 as their Bf6; transferred move h8f6. Synthesis: 1 retrieved analog transferred to Qxf8+ (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



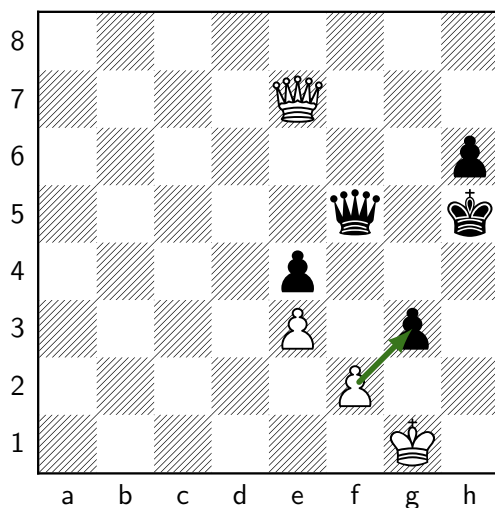
(a)



(b)

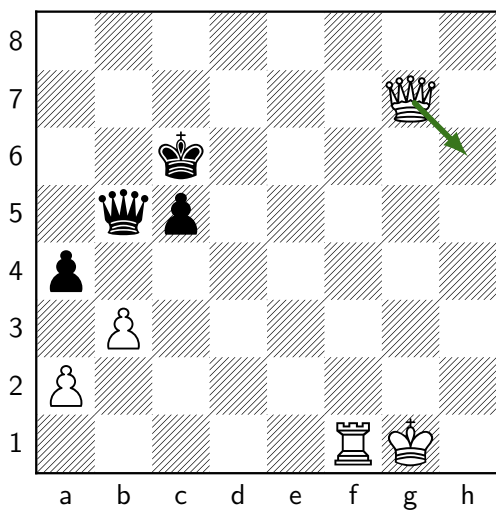


(c)

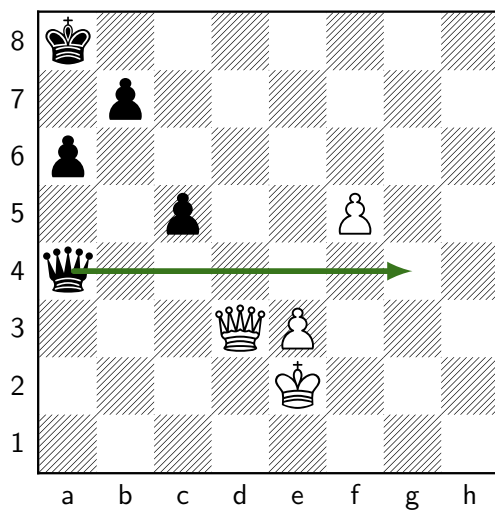


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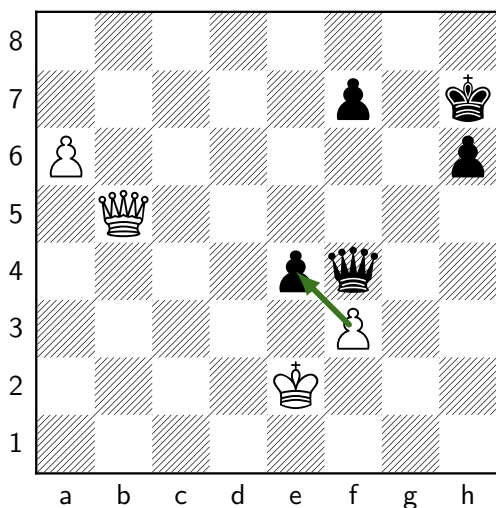
Figure 40: **Move 46, White: Qg7+.** (a) Our position; Melanie plays Qg7+ (f8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2897` (ply 83) [structural distance $d=45$, QAP obj=18]: the master played Qb2; under μ our Qf8 plays the role of that game’s Qc6; cross-type roles: our Rf1 as their Pf7; transferred move f8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2798` (ply 80) [structural distance $d=39$, QAP obj=84]: the master played gxf3; under μ our Pb3 plays the role of that game’s Pg2; cross-type roles: our Qf8 as their Pc3, our Rf1 as their Ph3; transferred move b3a4. (d) Retrieved from `ccr1_top_p0002#392` (ply 104) [structural distance $d=36$, QAP obj=55]: the master played fxf3; under μ our Pb3 plays the role of that game’s Pf2; cross-type roles: our Rf1 as their ph6; transferred move b3a4. Synthesis: 1 retrieved analog transferred to Qg7+ (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



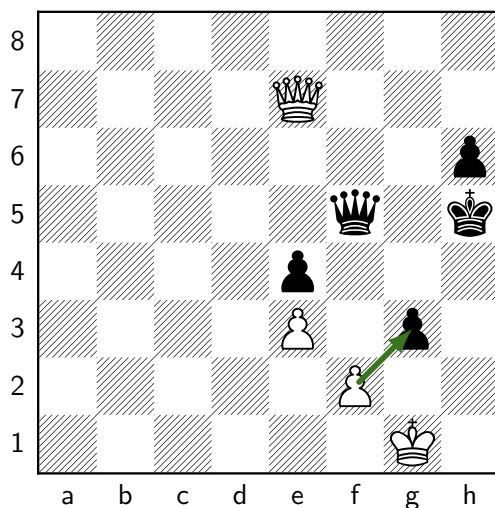
(a)



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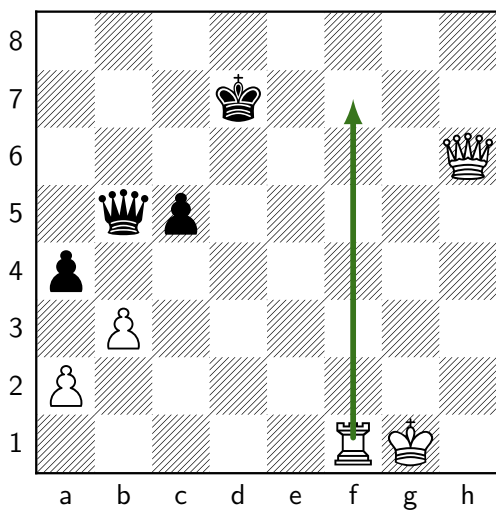


(c)

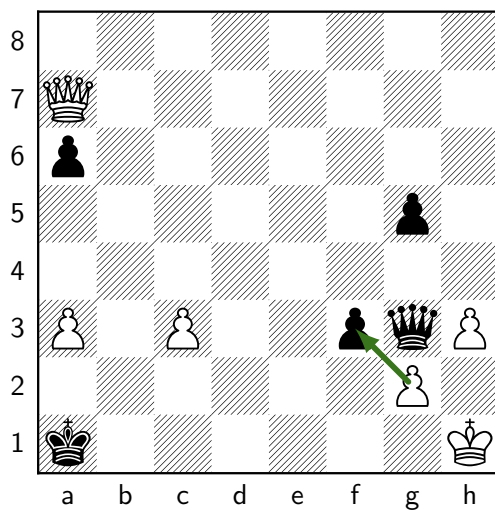


(d)

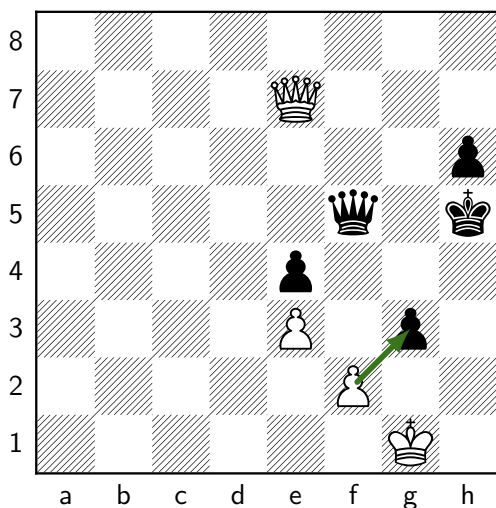
Figure 41: **Move 47, White: Qh6+.** (a) Our position; Melanie plays **Qh6+** (g7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1938` (ply 81) [structural distance $d=40$, QAP obj=9]: the master played **Qg4+**; under μ our *Qg7* plays the role of that game's *Qa5*; cross-type roles: our *Rf1* as their *Pa6*; transferred move **g7h6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#426` (ply 106) [structural distance $d=32$, QAP obj=64]: the master played **fxe4**; under μ our *Pb3* plays the role of that game's *Pf3*; cross-type roles: our *Rf1* as their *Qb5*, our *Qg7* as their *pf7*; transferred move **b3a4**. (d) Retrieved from `ccr1_top_p0002#392` (ply 104) [structural distance $d=34$, QAP obj=50]: the master played **fxg3**; under μ our *Pb3* plays the role of that game's *Pf2*; cross-type roles: our *Rf1* as their *ph6*; transferred move **b3a4**. Synthesis: 1 retrieved analog transferred to **Qh6+** (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



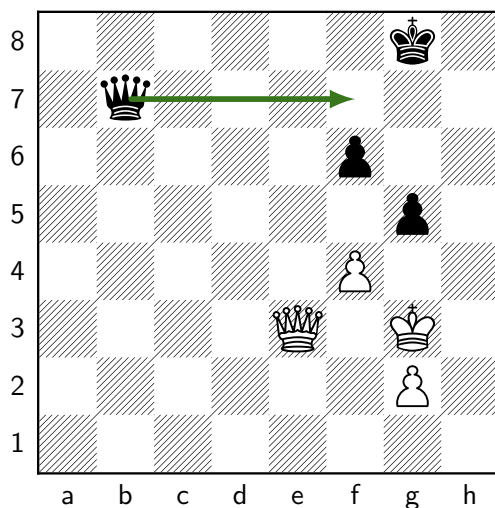
(a)



(b)

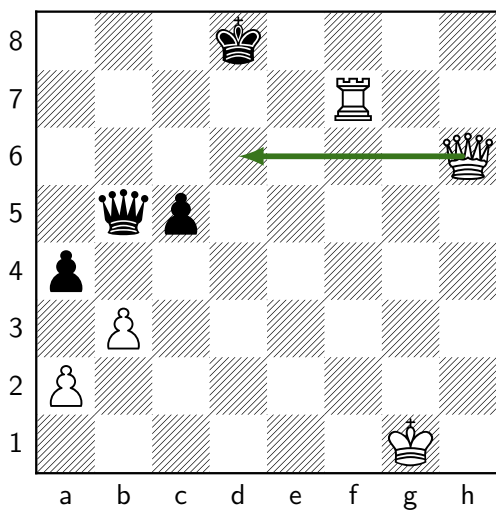


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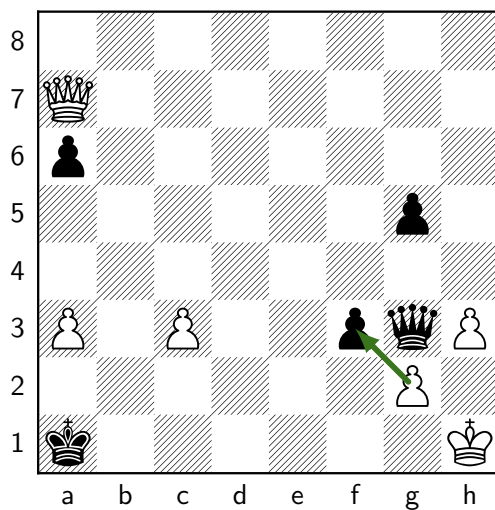


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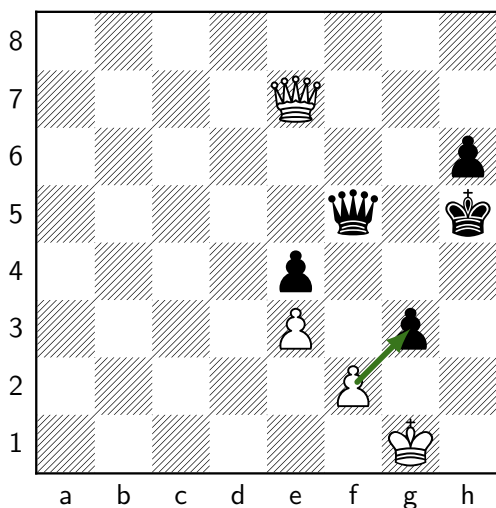
Figure 42: **Move 48, White: Rf7+.** (a) Our position; Melanie plays Rf7+ (f1f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#2798` (ply 80) [structural distance $d=27$, QAP obj=84]: the master played gxf3; under μ our *Pb3* plays the role of that game’s *Pg2*; cross-type roles: our *Qh6* as their *Pc3*, our *Rf1* as their *Ph3*; transferred move b3b4. (c) Retrieved from `ccr1_top-p0002#392` (ply 104) [structural distance $d=24$, QAP obj=49]: the master played fxg3; under μ our *Pb3* plays the role of that game’s *Pf2*; cross-type roles: our *Rf1* as their *ph6*; transferred move b3a4. (d) Retrieved from `ccr1_top-p0002#1502` (ply 109) [structural distance $d=30$, QAP obj=35]: the master played Qf7; under μ our *Qh6* plays the role of that game’s *Qb2*; cross-type roles: our *Rf1* as their *pg2*; transferred move h6h2. Synthesis: no retrieved analog transferred to Rf7+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



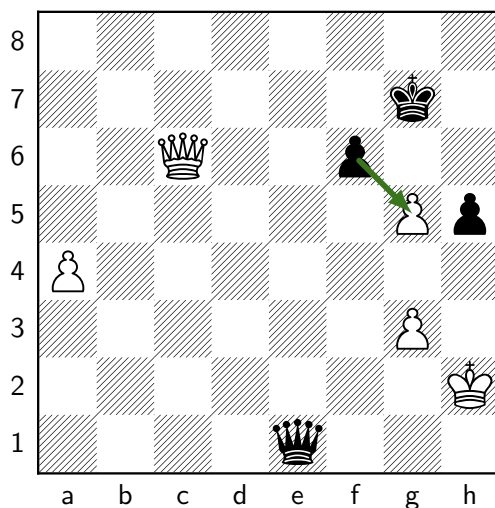
(a)



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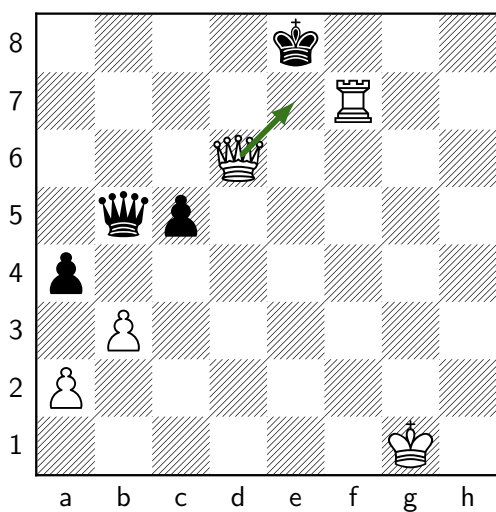


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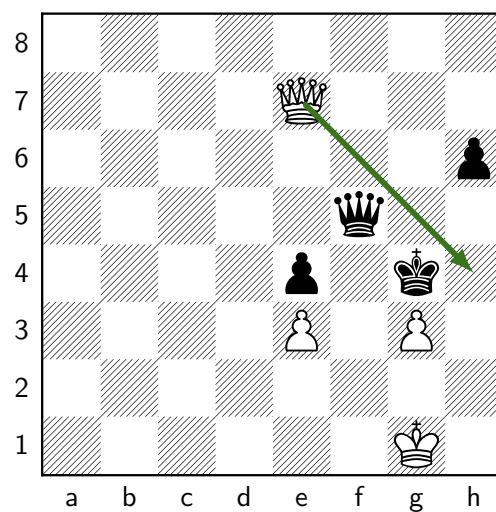


(d)

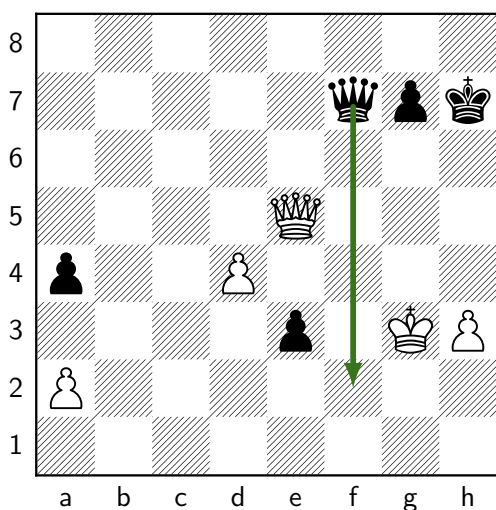
Figure 43: **Move 49, White: Qd6+.** (a) Our position; Melanie plays Qd6+ (h6d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#2798` (ply 80) [structural distance $d=23$, QAP obj=92]: the master played `gxf3`; under μ our *Pb3* plays the role of that game’s *Pg2*; cross-type roles: our *Qh6* as their *Pc3*, our *Pb3* as their *Qa7*; transferred move `b3b4`. (c) Retrieved from `ccr1_top-p0002#392` (ply 104) [structural distance $d=23$, QAP obj=56]: the master played `fxg3`; under μ our *Pb3* plays the role of that game’s *Pf2*; cross-type roles: our *Qh6* as their *ph6*, our *Rf7* as their *Qe7*; transferred move `b3a4`. (d) Retrieved from `ccr1_top-p0000#98` (ply 97) [structural distance $d=25$, QAP obj=50]: the master played `fxg5`; under μ our *Pb3* plays the role of that game’s *Pf3*; cross-type roles: our *Pb3* as their *Qe1*; transferred move `b3a4`. Synthesis: no retrieved analog transferred to Qd6+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



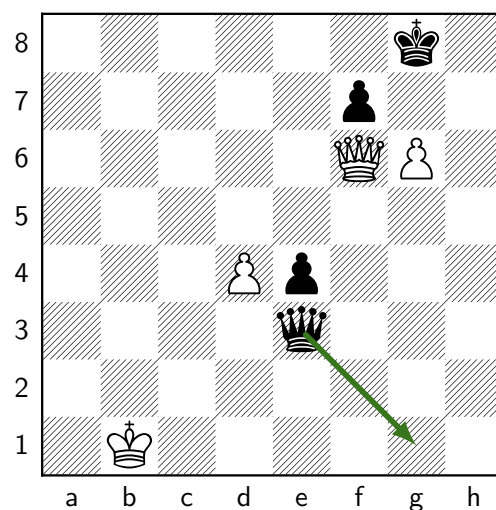
(a)



(b)



(c)



(d)

Figure 44: **Move 50, White: Qe7#.** (a) Our position; Melanie plays Qe7# (d6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#392` (ply 106) [structural distance $d=33$, QAP obj=5]: the master played Qh4+; under μ our Qd6 plays the role of that game's Qe7; cross-type roles: our Rf7 as their Pg3; transferred move d6e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3522` (ply 105) [structural distance $d=35$, QAP obj=4]: the master played Qf2+; under μ our Qd6 plays the role of that game's Qf2; cross-type roles: our Rf7 as their Pg7; transferred move d6e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2038` (ply 115) [structural distance $d=26$, QAP obj=20]: the master played Qg1+; under μ our Qd6 plays the role of that game's Qe6; transferred move d6b8. Synthesis: 2 retrieved analogs transferred to Qe7# (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.